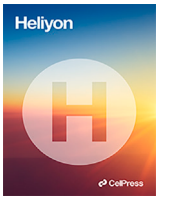
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Research article



State-of-the-art and real-time implementation of an IoT-based home energy management system for a cluster of dwellings

Nikita Ramachandra, Rajasekar Natarajan *

Solar Energy Research Cell (SERC), School of Electrical Engineering, Vellore Institute Technology, Vellore, Tamil Nadu-632014, India

A R T I C L E I N F O

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A B S T R A C T

In the present day electricity demand, demand response programs support mitigating the power demand and help to improve stability. Within this framework, the Home Energy Management System (HEMS) plays a critical role in optimizing energy consumption patterns by redistributing loads from peak to off-peak hours, thereby subsequently contributing to grid stability. The existing HEMS model often fails to simultaneously address the three important issues. 1. Minimizing power bills; 2. Maintaining peak-to-average ratio (PAR); and 3. User convenience while load scheduling. These challenges are further compounded by limitations like slow convergence in the existing optimization technique. Moreover, the lack of real-world validation impedes demonstrating their effectiveness and practicality for adoption. Thus, addressing these issues, this paper proposes the real-time implementation of the results of the optimization technique via a smart plug and a principal load scheduler (PSC) supported by a mobile application. First, the data prevalent to optimization techniques is collected from the stakeholders, and secondly, the multi-objective mountain gazelle optimization (MMGO) algorithm is formulated and utilized by the PSC to schedule household applications for all the individuals within the cluster. In addition, results are validated via a hardware prototype using a smart plug. Further, the suggested method is contrasted with existing multi-objective techniques and weighted techniques to demonstrate its superiority. While tested for single dwellings, the proposed method achieves a reduction of 12.14% in electricity costs and 52.54% in PAR compared to the unscheduled loads. Notably, during peak hours, it achieves a reduction of up to 80.15% in electricity costs and a 25.07% reduction in PAR. Extending the analysis further to multiple dwellings, 50 homes in a cluster, reveals an overall cost reduction and PAR reduction of 11% and 74.68%. Additionally, the assimilation of PV systems and battery management systems into the smart home would result in lucrative benefits and a flattening of the net load demand curve.

Nomenclature

Abbreviations

$BESS$	Battery energy storage system
$CCHP$	Combined cooling, heating, and power
CSA	Cuckoo optimization algorithm
DR	Demand response
$GWCSO$	Grey wolf and crow search optimization
$HEMS$	Home energy management systems

* Corresponding author.
E-mail address: nrajasekar@vit.ac.in (R. Natarajan).

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IBR	Inclination block rate
IoT	Internet of things
$MAOA$	Multi-objective arithmetic optimization algorithm
$MILP$	Mixed integer linear programming
$MMGO$	Multi-objective mountain gazelle optimization
$NSGA-II$	Non-dominated sorting genetic algorithm-II
PAR	Peak-to-average ratio
PEV	Plug-in electric vehicle
PSC	Principal scheduling controller
PSO	Particle swarm optimization
RES	Renewable energy source
RTP	Real-time pricing
TOU	Time-of-Use
CF	Net cashflow
CPP	Cash payback period
MFF	Multi-objective Firefly
OMB	Operation and maintenance bills
TMH	Territorial solitary males
TNEB	Tamil Nadu Electricity Board

Variables

$\tau_{begin_{h,d_{ul}}}$	User-defined start time of the appliance
$\tau_{end_{h,d_{ul}}}$	User-defined end time of the appliance
$\tau_{sch_{d_{ul}}}$	Appliance start time suggested by PSC
$\tau_{span_{h,d_{ul}}}$	Operational time of appliance
$\tau_{wait_{d_{ul}}}$	Appliance waiting time
Ar_{pvm}	Surface area of PV module
$bound_{lower}$	Lower limit of objective function
$bound_{upper}$	Upper limit of objective function
C^{τ}_{TOU}	TOU tariff signal for time slot τ
CC_{RES}	Installation cost of RES
$CC_{Smart\ Plug}$	Installation cost of smart plugs
$Coef_n$	Coefficient vector
CC_{BESS}	Installation cost of BESS
D	Total appliances in a home
D_{rl}	Regular appliances
D_{ul}	Uninterruptible-time-deferrable appliances
DoD_{bt}	Depth of discharge
E^{τ}_{bch}	Battery charging energy at time instant τ
E^{τ}_{bdch}	Battery discharging energy at time instant τ
$E^{\tau}_{bt_{max}}$	Maximum battery capacity
E^{τ}_{gbuy}	Energy consumed from the grid
E^{τ}_{gsell}	Energy sold to the grid
$E^{\tau}_{MAXgbuy}$	Maximum energy allowed to purchase from grid
$E^{\tau}_{MAXgsell}$	Maximum energy allowed to sell to grid
E^{τ}_{rl}	Total energy consumption of all regular appliance at time slot $^{\tau}$
E^{τ}_{ul}	Total energy consumption of all uninterruptible-time-deferrable appliance
gaz_{male}	Adult male gazelle
$gaz_{male_{child}}$	Young male herd coefficient vector
H	Total dwellings in a cluster
It	Current number of iteration
M_It	Total number of iterations
NPV_{HEMS}	Net present value of the HEMS system
P^{τ}_{bch}	Power consumed while charging battery
P^{τ}_{bdch}	Power drained while discharging battery
$rand$	Uniform random vector (0 to 1)
rd	Random integer of value (1 or 2)
SE^{τ}_{pvm}	Energy generated by PV module at time instant τ
SoC	State of charge
SE_{total}	Total solar energy produced
T	Total time slots
T^{τ}_{o}	Ambient temperature
$Y(t)$	Position of the gazelle vector in the current iteration
Y_r	Total number of years
B_z	Random number from the standard distribution
H_{pop}	Fitness matrix
$S_{h,d_{rl}}$	Power consumption of each regular appliance
$S_{h,d_{rl}}$	Power consumption of each uninterruptible-time-deferrable appliance
Y_{max}	Lower limit of Search space
Y_{min}	Upper limit of Search space
Y_{nd}	Search space
η_{bch}	Charging rate of battery
η_{bdch}	Discharging rate of battery
η_{pvm}	Efficiency of PV module

1. Introduction

Compelled work-from-home regulations after COVID-19 pandemic, plug-in electric vehicles (PEVs) in residence, use of battery energy storage systems (BESS) and rapid urbanization increased energy consumption in India from 8,24,301 GWh to 12,963,300 GWh between 2012 and 2022, out of which the domestic sector accounted for a share of 25.77% [1]. Therefore, the participation of residential users in demand-side management (DSM) programs enacted by utilities can sustain power quality, obviate deploying additional power units and spinning reserves, and reduce power outages. The key components of the program involve energy efficiency management, energy storage management, and the demand response (DR) program [2]. Energy efficiency management can be achieved by employing energy-star-certified appliances in dwellings. Furthermore, the energy storage management program encourages consumers to become prosumers by producing, storing, and distributing energy using renewable energy sources (RES) and battery energy storage systems (BESS).

The implementation of the DR program introduces an advanced energy pricing structure with the aim of accurately reflecting production costs for customers. This approach encourages users to voluntarily shift loads from peak times to off-peak hours in exchange for monetary reductions. Advanced energy pricing schemes can be either price-based, where high costs dissuade power usage and low prices promote it, or incentive-based, where monetary reimbursement is provided for reducing power consumption over a certain time period. In practical applications, the price-based DR program is usually more suited for residential customers than the incentive-based DR program [3].

Under such circumstances consumers can either manually alter their usage or use home energy management systems (HEMS) to automatically charge BESS while electricity prices are low, detect the intervals of RES generation, and react to energy price changes received by utilities. Therefore, an internet of things (IoT)-based bidirectional communication link is established between HEMS and

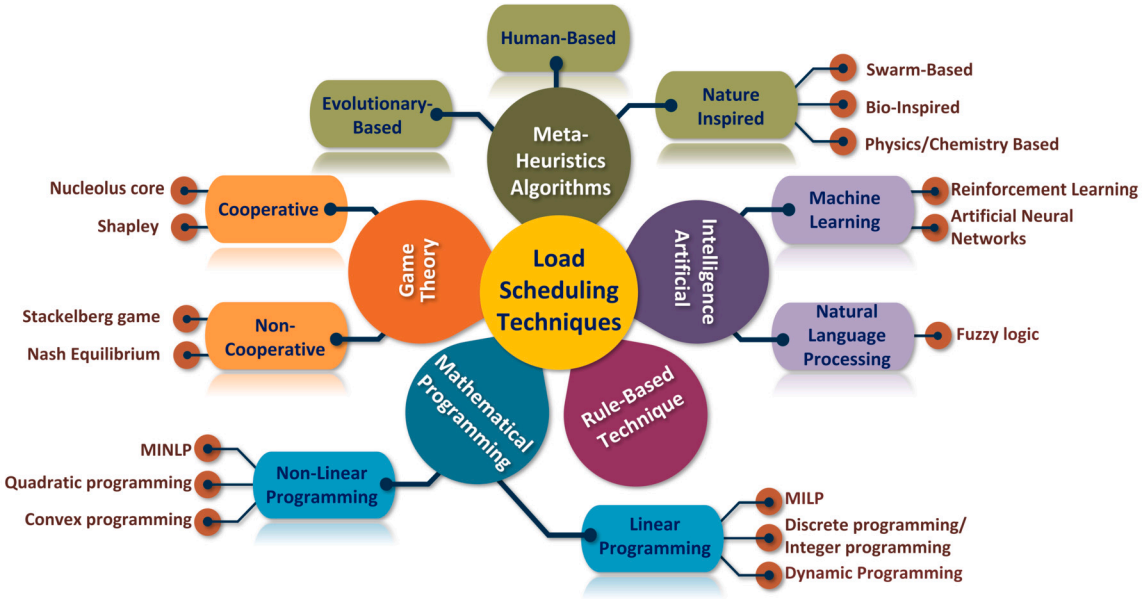


Fig. 1. Different optimization schemes applied in demand response program.

the grid-connected smart meter to provide connectivity across utilities and prosumers. Some HEMS controllers use PV forecast data to prioritize charging among BESS and load, while others employ actual data. However, PV forecasts are achieved using traditional meteorological techniques, satellite photos, or ground-based sky imaging data. Since these data are frequently provided by third-party services, it incurs a supplement cost to the system.

HEMS automates load scheduling by incorporating various optimization techniques, including mathematical programming, rule-based techniques, artificial intelligence, game theory, and meta-heuristics algorithms, as illustrated in Fig. 1. The optimization is performed while adhering to constraints like electricity prices, prosumer behavior, RES and BESS energy, user comfort, and the peak-to-average ratio (PAR) of energy demand. Several well-known mathematical programming approaches, including quadratic programming, convex programming, and mixed-integer linear programming, are mentioned in the literature [4], [5], [6], and [7], although this technique suffers from a slow convergence rate for large-sized problems. Alternately, the rule-based approach outlined in [8], [9], and [10] falls short as the system is broadened. Whereas the use of artificial intelligence approaches, as detailed in [11], [12], [13], and [14], necessitates extensive device assessment for validation and affirmation. Despite the intricacy, game theory as presented in the literature [15], [16], and [17] is challenging to model and implement in real-world scenarios, even though it operates efficaciously in decision-making under uncertain conditions. Besides, meta-heuristic algorithms have a fast convergence speed with the potential to resist premature convergence and can solve non-linear and extremely complicated multi-dimensional optimization problems. The implementation of meta-heuristic algorithms in HEMS has been documented in various literatures. These include particle swarm optimization (PSO) [18], genetic algorithm [19] [20], bi-level particle swarm optimization and evolutionary algorithm [21], multi-objective firefly algorithm (MFF) [22] [23], hybrid grey wolf optimizer with min-conflict algorithm [24], hybrid harmony search algorithm with differential evolution [25], mutation operator integrated ant colony optimization [26], fractional-order future search optimizer [27] and genetic binary particle swarm optimization [28]. Nonetheless, the convergence performance and computational time of these approaches are modest. As a result, a multi-objective mountain gazelle optimization (MMGO) is proposed for HEMS, which operates under the Time-of-Use (TOU) pricing structure and significantly reduces the PAR with improved end-user convenience while lowering power costs. A photovoltaic (PV) system and BESS are integrated with HEMS to further lower the cost of power without adopting a PV forecast.

2. Literature review and contributions

A plethora of research studies have been carried out in the literature on several aspects of developing HEMS models that incorporate network frameworks, pricing schemes, user comfort, and multi-objectivity solving using various optimization techniques.

Ref. [29] presented a HEMS model based on the mixed integer linear programming (MILP) architecture and explored the effects of various tariff schemes on demand management. Therefore, it is determined that the real-time pricing (RTP) structure mirrored the actual market conditions, whereas the TOU tariff gave consumers a choice to reduce the bill amount beforehand. For the purpose of further cost reduction and enabling energy exchange with the utility grid, RES and BESS were taken into consideration. However, the drawback of the model is that the operating durations are specified, which might vary in real-world situations, and the reduction of PAR is also not considered.

The authors of [30] developed a pre-emptive priority-based load scheduling algorithm for HEMS that operates in accordance with TOU pricing tariffs. Consequently, the prioritization of appliance scheduling led to a reduction in electricity bills, and the

incorporation of PV and BESS lowered peak demand. Although the main goal of minimizing electricity bills is achieved, concerns like user comfort and PAR are overlooked.

In [31], the study focuses on the design and evaluation of a HEMS model for active, semi-active, and fully active residential users. The primary objective of this system is to identify the optimal time period to schedule appliances in order to minimize user discomfort, power consumption costs, and PAR through a sine-cosine optimization algorithm. It is reported that the model reduces the energy consumption of multiple residences at peak times.

A bilevel multi-objective optimization strategy has been developed in [32], where initially an enhanced normalized normal constraint approach employing game theory is applied. Then a fuzzy compromising method is applied to find the optimal solution from the Pareto optimal front. The method is compared under the RTP scheme with seven other methods to demonstrate its supremacy. Furthermore, the PV and wind generation forecasting models are used and integrated with combined cooling, heating, and power (CCHP) systems, BESS, and thermal energy storage systems. However, the approach has not been validated for multiple smart homes.

Ref. [33] uses a hybrid grey wolf and crow search optimization (GWCSO) algorithm to schedule consumer electronics within the real-time price signals (RTPS) tariff structure while accounting for power cost, PAR, and user comfort. In an effort to demonstrate the effectiveness of the algorithm, it is compared to seven other algorithms. The suggested algorithm reduced peak power by 5.16%, 16%, 19.45%, 13.01%, 16%, and 20.54% in comparison to crow search, grey wolf, firefly, harmony search, earthworm, bacterial foraging, and genetic optimization techniques.

In [34], the load scheduling issue is presented as a weighted objective function and is solved using an optimization model based on binary integer linear programming. However, the weighting factor adversely affects the user satisfaction metric. The presented model, when compared with mixed-integer non-linear programming, minimized computational time and cut electricity costs by 31%. Further experimental findings revealed that for the small-scale and large-scale case studies, the given optimization model reduced power expenditures by around 35% and 50% in contrast to the unscheduled load. However, the authors did not delve into the argument for integrating RES and BESS.

The authors of [35] utilized non-dominated sorting genetic algorithm II (NSGA-II), which aims to reduce power consumption costs and PAR while considering resident convenience. It is noticed that significant power demand is averted during periods of low cost by employing an RTP together with an inclination block rate (IBR) tariff, resulting in a PAR value of 2.04 and a daily cost value of 87.2 cents. Thus, the installation of a BESS system decreased the PAR and daily cost value to 1.893 and 86.08 cents, respectively. However, the researchers did not carry out the validation of the HEMS model for multiple dwellings.

To reduce the peak load and power cost of a smart house, [36] presented a MILP-based HEMS model integrated with RES and BESS. The impact of China-based PV subsidy programs from 2013 to 2018 on the optimization of home appliances is also examined. It has been noticed that the cost of electricity in 2013 was 60% cheaper than in 2018. Nevertheless, the cost of PV technology has also declined throughout the years.

The authors of [37] defined HEMS objectives as a weighted sum MILP approach for multiple residential consumers with the BESS. The objectives involved electricity bill reduction and peak load minimization; however, user convenience was disregarded. It is deduced through simulation analysis that the HEM paired with BESS reduces annual power bills for single and multiple households by 573.05 and 565.75 dollars under the ToU tariff plan, respectively.

The authors of [38] have devised a more efficient cooperative PSO with stochastic attraction-repulsion of diversity to address the load scheduling issue while lowering user discomfort and energy costs. The method exhibits a better convergence rate in comparison to seven other established cooperative PSO-based methods. Moreover, in accordance with a case study on HEMS for various housing units, practically every resident saved 1,226.4 euros on power expenses with coordination and 722.7 euros without coordination annually.

The authors of [39] proposed a novel Pareto tribe evolution for multiple HEMS with an equilibrium-based decision. The nash equilibrium-based decision-making identifies the ideal compromise solution for three objective functions: minimization of energy cost, PAR, and user comfort of the load profile from the Pareto optimal front. Subsequently, the outcomes are contrasted with five multi-objective algorithms, and it is determined that the recommended strategy surpasses other algorithms in every aspect. The approach is evaluated using 100 HEMS models under the TOU pricing scheme, but the integration of RES and BESS is not considered.

In [40], RES and BESS are combined with a multiobjective HEMS model. The goal of this model is to minimize both the cost of electricity and the change in the load profile by using linearization techniques. For the simulation study, three case scenarios are considered where HEMS with RES and BESS decreased the electricity cost by roughly 23% and the load profile deviation by 46% in comparison to HEMS with RES. Whereas, adding RES, BESS, and PEVs to HEMS lowered electricity costs and load profile deviation by 7% and 62%, respectively, in contrast to HEMS with RES and BESS.

Ref. [41] presents HEMS, which employs MILP to alleviate flexible load during peak hours with the aim of minimizing electricity costs and user comfort. According to the simulation study, the suggested HEMS model reduced electricity expenses by 36.0% during the winter and 33.3% during the summer. Additionally, the HEMS model gave residential consumers choice by allowing them to choose between three categories based on their preferences: comfort, economics, and neutrality.

To optimize the behavior of home appliances with respect to retail energy costs, a novel HEMS model based on the forward-backward heuristic algorithm and linear programming is put forth in [42]. In addition, stochastic programming is used to deliver the hourly retail pricing from the merchant to the HEMS in order to minimize the weighted function comprising of operation costs, PAR, and load profile congestion. The issue of load shedding was alleviated, and the average retail price over a 24-hour period dropped by 1.5% after HEMS was implemented, whereas without HEMS, the congestion rate was approximately 37%. However, throughout HEMS modeling, the consumer satisfaction metric was disregarded.

The optimization model in [43] is designed as a two-stage stochastic problem and is formulated as a MILP problem that intends to minimize the estimated average cost of a consumer. Additionally, first-order markov chain and multi-layer perceptron neural networks are employed to anticipate solar radiation and wind speed in order to calculate the forecast of RES. It is observed that the HEMS model reduced consumer costs and PAR under the combination of RTP and IBR pricing schemes by allowing household appliances to be shifted according to expected RES output and electricity pricing. But the customer discomfort index is not factored in while establishing the objective function. Additionally, it is demonstrated that participation in the contractual market has elevated the amount of energy sold to the network due to the concurrent reduction in net costs and purchase expenses.

NSGA-II is employed in [44] to compute compromise solutions by balancing the minimization of total cost and user dissatisfaction. By using contracted power and peak power charges, a case study was carried out on four distinct types of loads including shiftable loads, thermostatically controlled loads, BESS, and PEVs. The findings demonstrate that peak power pricing models outperform contracted power pricing models as they encourage reduced energy expenses and peak power. Additionally, the HEMS model is implemented using Java programming on the Raspberry Pi 3 Model B+.

Ref. [45] introduces the PSO and binary PSO hybrid heuristic algorithm to address the load scheduling issue by establishing a balance between user convenience and electricity costs. Additional factors employed in load scheduling include historical consumer preference patterns and PV forecasted data, which are produced by machine learning clustering algorithms utilising the K-Means approach and an auto-regressive integrated moving average PV prediction, respectively. In addition, BESS and PEVs are leveraged to reduce the electricity bill under the TOU pricing scheme. The HEMS model is developed using a Raspberry Pi microcontroller and tested on real data acquired from a smart house. The model executed 63.15% faster and is capable of lowering power costs, PAR, and energy consumption by 52.92%, 44%, and 69.44%, respectively, when compared to the PSO technique.

In [46], a support vector regression-based artificial neural network is utilised to forecast wind and PV power output. HEMS uses the predicted output to shift the load, utilising reformed optimization based on NSGA-II. The power cost, the user's priority for using their appliances, and BESS are accounted for while scheduling loads. As an outcome, under the TOU pricing plan, the HEMS model is able to lower power bills by 51.4%. However, the model is not assessed on a group of households, and the effect on the PAR metric is also not examined.

Ref [47] implemented a PSO approach to shift loads depending on the day-ahead pricing scheme as well as RES and BESS power production. The approach is tested on 12 dwellings, where load scheduling performed using RES solely and integrating RES with BESS resulted in reductions in energy costs of 22% and 45%, respectively. But PAR and comfort levels of occupant are ignored.

An improved enhanced differential evolution algorithm is developed in [48] to execute load shifting while minimizing power costs, PAR, user discomfort, and uncertainty in PV production. Moreover, in comparison to improved differential evolution, NSGA-III, and dominance and decomposition based multi-objective evolutionary algorithm, the suggested approach required less computing time. Further, when compared to an unscheduled home, the PAR decreased by 41.7%.

The authors of [49] scrutinized a BESS-integrated HEMS model for load scheduling under RTP and TOU rates. A multi-objective MILP is employed to schedule the HEMS model with the intention of reducing energy consumption and consumer discomfort. The simulation analysis is carried out for HEMS with and without BESS under two pricing schemes. HEMS without BESS resulted in a reduction of electricity costs by 32.32% and 11.22% under TOU and RTP tariffs, respectively. Conversely, the installation of BESS resulted in 35.35% and 14.7% electricity cost reductions under the TOU and RTP tariffs, respectively. However, the PAR metric is not considered during evaluation.

A binary particle swarm optimization technique with a quadratic transfer function is suggested in [50]. The approach allows shiftable appliances to be scheduled under RTP and TOU rates to lower power costs and consumer discomfort. The integration of RES and BES, however, is not taken into account. When compared to the conventional binary PSO and nine other binary PSO versions, it is reported that the bill reduction for the binary PSO with quadratic transfer function is as high as 50.76%.

A limited memory method is employed in [51] to minimize costs and user inconveniences when load shifting from peak to non-peak hours under TOU pricing is performed. In order to further lower the electricity expenditure, the HEMS model shifts the functioning of some appliances when PV power is available. As a result, customers save 32.93% on their daily electricity bills.

The TOU power tariff is used in [52] to schedule appliances, with the electricity cost, user discomfort, and the PAR as the objective functions. The addition of RES and BESS, therefore, would have assisted in further reducing the load during peak hours. When compared to the three existing MILP models described in the literature, it is discovered that the recommended approach lowers electricity costs, user discomfort, and PAR. In further analysis under the multiple-household scenario, the PAR dramatically decreased, leading to better load profiles.

Appliance scheduling utilizing the beetle antennae search algorithm is done in [53] using price-based and incentive-based demand response schemes. To further assure a decrease in power costs, RES and BESS are incorporated.

In [54], a modified butterfly algorithm is employed to develop a HEMS model that is implemented on ZigBee while taking user satisfaction and energy consumption costs into account. When compared to PSO-based and BOA-based algorithms, the findings were effective.

In [55], a multi-objective arithmetic optimization algorithm (MAOA) is used to solve a tri-objective function under real-time pricing and critical peak pricing scheme. The NodeMCU and Node-RED modules of the Raspberry Pi minicomputer are used to setup the HEMS, which allows the scheduling of devices by tracking parameters such as voltage, current, power, and momentary energy consumption. However, in contradistinction to the antlion optimization, PSO, and grey wolf optimization algorithms, the MAOA approximately reduces daily electricity bills by 30%.

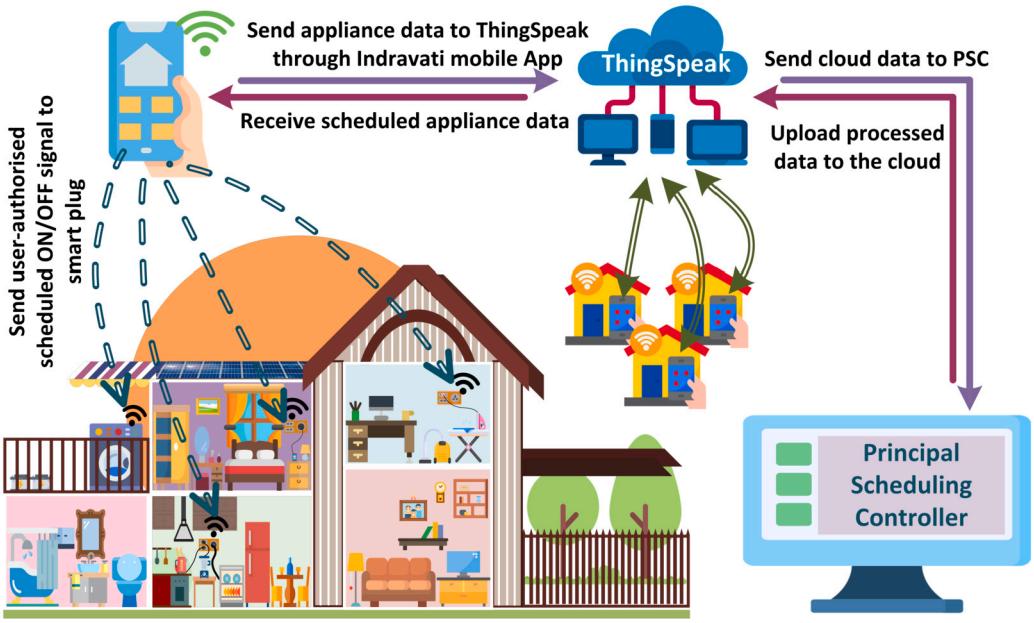


Fig. 2. Structural elements of the proposed IoT-based smart home.

The cuckoo optimization algorithm (CSA) [56] is used to solve the tri-objective function using a weighted approach. The effectiveness of the Cuckoo algorithm is validated by reviewing it with the MILP model. Further, the research is implemented in real-time on an academic building in Egypt, and the results indicate that the institution saved between 57% and 80% on energy bills.

Based on the comprehensive literature conducted heretofore (Table 1), it is evident that the preponderance of research has not acknowledged all three objectives (minimization of the power bill, PAR, and user inconvenience) concurrently. Whereas some literature has used a weighted sum approach to implement multi-objective functions as a single objective function. The shortcomings of weighted functions are that the optimal solution is highly influenced by the weights associated with each function. Further, the weighting factors are determined by applying the trial-and-error procedure without any assurance of achieving Pareto optimal values for these factors. Additionally, load scheduling for clusters of dwellings is not considered, which might lead to higher PAR during off-peak hours since each home likely prefers to operate its appliances when power prices are low. Perhaps to fill this research gap, the primary contributions of the present study are succinctly abridged as follows:

- The issue of demand scheduling is reformulated in the context of a multi-objective optimization model for manifold dwellings. A novel multi-objective mountain gazelle optimization (MMGO) is devised to strategize loads for energy management in the residential sector within the constraints.
- The proposed scheduling approach minimizes PAR, which alleviates the eventuality of blackouts and the entailment of additional plants, curtailing operating costs and benefiting the utility.
- The residents obtain a monetary reduction in energy bills by making minimal compromises while powering the appliance with the ToU price-based demand response technique.
- Domestic users are subjected to a price-based DR scheme with the amalgamation of PV systems and BESS to pare down the peak demand of the main grid.
- Furthermore, to authenticate the technique in the real-world scenario, a prototype of a smart plug, a principal load scheduler, and a mobile application are fabricated and appraised.

3. System architecture and its operation strategy

In this study, the authors developed a HEMS model for clusters of smart homes, where each home is powered by the grid, a PV array, or a BESS. The proposed HEMS includes a principal scheduling controller (PSC), smart plugs, and a mobile application as shown in Fig. 2. Subsequently, all data is collected and transmitted to the PSC controller, which schedules the residential loads within a cluster. The communication and interaction between these components are described as follows:

3.1. Principal scheduling controller

The PSC acts as the central hub for all HEMS in the cluster. Users notify the PSC of their preferred operating times for each appliance. The PSC receives pricing data from the utility and combines it with the user-provided appliance data to perform scheduling.

Table 1
Comprehensive literature analysis pertaining to the load scheduling model implementation methods.

Optimization Techniques	Methodology Employed	Objective Functions	Constraints	Tarrif Scheme	Techniques used for comparison	Remarks
Mathematical programming	MILP [29]	• Cost minimization. • User comfort maximization. • Start-up and the shut-down costs of the assets.	• PV system, BESS, load and grid power constraints. • Charging/discharging efficiency and state of charge (SoC) of the battery. • Load scheduling period.	• RTP • TOU • IBR	N.A.	• PAR reduction is not considered. • Model implication for multiple homes is neglected.
Rule-based technique	Pre-emptive priority-based load scheduling technique [30]	• Cost minimization.	• PV system, BESS, load and grid power constraints. • Battery SoC. • Load scheduling period. • Temperature restriction	• TOU	N.A.	• PAR reduction is not considered.
Meta-heuristic algorithm	Sine cosine algorithm [31]	• Cost minimization. • User comfort maximization. • PAR minimization.	• PV system, BESS, load and grid power constraints. • Maximum/minimum energy and SoC of the battery. • Load scheduling period.	• TOU • IBR	• Genetic Algorithm • Particle swarm optimization • Whale optimization	• Formulated objective function as weighted function.
Game theory	Game theory unified enhanced normalized normal constraint [32]	• Cost minimization. • Gaseous emission minimization • Reluctance to take risks.	• PV and wind, CCHP, thermal energy storage, BESS, load and grid power constraints. • Charging and discharging rate of battery. • Maintenance costs of wind and PV system, CCHP generator, BESS and thermal energy storage system. • Load scheduling period. • Temperature restriction	• RTP	• Particle swarm optimization • Grass-hopper optimization • Salp-swarm optimization • Flower pollination optimization • Coyote optimization • Sine–cosine optimization • Genetic optimization • Grey wolf optimization • Tabu search optimization	• Model implication for multiple homes is neglected.
Meta-heuristic algorithm	Grey wolf and crow search optimization [33]	• Cost minimization. • User comfort maximization.	• Minimum/maximum PAR. • Load and grid power constraints. • Load scheduling period.	• RTP	• Genetic Algorithm • Bacterial foraging optimization • Earthworm optimization • Harmony search algorithm • Firefly optimization • Crow search optimization • Grey wolf optimization	• Formulated objective function as weighted function. • Model implication for multiple homes is neglected. • Integration of RES and BESS is lacking.
Mathematical programming	Binary integer linear programming [34]	• Cost minimization. • Incentives • User comfort maximization.	• Load scheduling period.	• TOU	• Mixed integer non-linear programming	• Integration of RES and BESS is lacking. • PAR reduction is not considered.
Meta-heuristic algorithm	Non-dominated sorting genetic algorithm-II [35]	• Cost minimization. • User comfort maximization.	• Minimum/maximum PAR. • Load scheduling period.	• RTP combined with IBR	N.A.	• Model implication for multiple homes is neglected.
Mathematical programming	Mixed integer programming [36]	• Cost minimization. • PAR minimization.	• Charging/discharging efficiency, energy, and SoC of the battery. • Load scheduling period.	• TOU	N.A.	• User discomfort minimization is not considered.
Mathematical programming	Multi-objective MILP [37]	• Cost minimization. • PAR minimization.	• Charging/discharging efficiency, energy, and SoC of the battery. • Load scheduling period.	• TOU	N.A.	• Formulated objective function as weighted function. • User discomfort minimization is not considered.
Meta-heuristic algorithm	Improved cooperative particle swarm optimization with stochastic attraction-repulsion of diversity [38]	• Cost minimization. • User comfort maximization.	• Load scheduling period. • Temperature restriction	• Time-varying	N.A.	• Integration of RES and BESS is lacking. • PAR reduction is not considered.

(*continued on next page*)

Table 1 (*continued*)

Optimization Techniques	Methodology Employed	Objective Functions	Constraints	Tarrif Scheme	Techniques used for comparison	Remarks
Meta-heuristic algorithm	Pareto tribe evolution with equilibrium-based decision [39]	• Cost minimization. • User comfort maximization. • PAR minimization.	• PV system, PEV, load and grid power constraints. • Charging/discharging efficiency, energy, and SoC of PEV. • Temperature restriction. • Load scheduling period.	• TOU	• Particle swarm optimization • Evolution algorithm based on decomposition • Non-dominated neighbor immune algorithm • Non-dominated sorting genetic algorithm	• Formulated objective function as weighted function.
Mathematical programming	Linearization techniques [40]	• Cost minimization. • PAR minimization.	• BESS and PEV conversion rates, charging/discharging state of energy. • Grid power constraints. • Load scheduling period.	• TOU	N.A.	• User discomfort minimization is not considered. • Model implication for multiple homes is neglected.
Mathematical programming	Mixed integer programming [41]	• Cost minimization. • User comfort maximization.	• Renewable energy system, CCHP, load and grid power constraints. • Charging/discharging efficiency, energy, minimum/maximum output power and SoC of PEV. • CCHP and water heater system heat energy balance constraint. • Temperature restriction of indoor air, water storage tank, CCHP. • Load scheduling period. • Illumination constraint. • Heating/cooling rate of air conditioner, chiller. • Load scheduling period.	• TOU	N.A.	• Formulated objective function as weighted function. • The designed model is tested in many circumstances, but not simultaneously for several residences.
Meta-heuristic algorithm	Forward-Backward Algorithm [42]	• Profit maximization of retailer. • Cost minimization. • Integrated PV and BESS payoff maximization	• Temperature restriction of indoor air, water storage tank, refrigerator. • Demand minimization of illumination system. • Charging/discharging efficiency, rate, power, and SoC of BESS. • Minimization of PAR and congestion.	• Retail electricity price	N.A.	• Formulated objective function as weighted function. • User discomfort minimization is not considered.
Mathematical programming	MILP [43]	• Cost minimization. • PAR minimization.	• PV and wind energy system, BESS, load and grid power constraints. • Power purchased/sold to grid. • Load scheduling period. • Charging/discharging power and SoC of BESS. • Illumination constraint.	• RTP • IBR	N.A.	• User discomfort minimization is not considered. • Model implication for multiple homes is neglected.
Meta-heuristic algorithm	Non-dominated sorting genetic algorithm-II [44]	• Cost minimization. • User comfort maximization.	• PV system, BESS, load and grid power constraints. • Temperature restriction of indoor air, water heater, refrigerator. • SoC of PEV. • Load scheduling period.	• TOU	N.A.	• PAR reduction is not considered.
Meta-heuristic algorithm	Hybrid particle swarm optimization and binary particle swarm optimization [45]	• Cost minimization. • User comfort maximization. • PAR minimization.	• Charging/discharging power and SoC of PEV and BESS. • Energy transfered to grid from PEV. • Load scheduling period.	• TOU	• Particle swarm optimization	• Shortfall in prototype development.
Meta-heuristic algorithm	Elitist non-dominated sorting genetic algorithm-II [46]	• Cost minimization. • User comfort maximization.	• Each day energy bill expense limit set by consumer. • Charging/discharging efficiency, energy and self discharge rate of BESS. • Load scheduling period.	• TOU	N.A.	• PAR reduction is not considered.

Table 1 (*continued*)

Optimization Techniques	Methodology Employed	Objective Functions	Constraints	Tarrif Scheme	Techniques used for comparison	Remarks
Meta-heuristic algorithm	Particle swarm optimization [47]	• Cost minimization.	• PV system, BESS, load and grid power constraints. • Charging/discharging power, efficiency, self discharge rate and SoC of BESS. • Converter efficiency. • Load scheduling period.	• Day ahead pricing	N.A.	• PAR reduction is not considered. • User discomfort minimization is not considered. • Model implication for multiple homes is neglected.
Meta-heuristic algorithm	Improved differential evolution algorithm [48]	• Incentive maximization by utilizing renewable source. • Cost minimization. • User comfort maximization. • PAR minimization. • Uncertainty of forecasted PV and load.	• Charging/discharging rate and SoC of BESS. • Load scheduling period.	• TOU	• Enhanced differential evolution • Multi-objective evolutionary algorithm based on dominance and decomposition • Non-dominated sorting genetic algorithm III	• Formulated objective function as convex function. • Model implication for multiple homes is neglected.
Mathematical programming	Multi-objective mixed integer linear programming [49]	• Cost minimization. • User comfort maximization.	• BESS, load and grid power constraints. • Charging/discharging efficiency, minimum/maximum energy stored in BESS. • Load scheduling period.	• RTP • TOU	• Dragonfly algorithm • Gravitational search algorithm • Backtracking search algorithm • Artificial bee colony • Particle swarm algorithm • Enhanced leader particle swarm optimization	• PAR reduction is not considered. • Model implication for multiple homes is neglected.
Meta-heuristic algorithm	Quadratic binary particle swarm optimization [50]	• Cost minimization. • User comfort maximization.	• Load scheduling period.	• RTP • TOU	• S-shaped and V-shaped transfer functions	• Integration of RES and BESS is lacking. • PAR reduction is not considered. • Formulated objective function as weighted function.
Mathematical programming	Mixed integer non-linear programming [51]	• Cost minimization.	• Load scheduling period.	• TOU	N.A.	• PAR reduction is not considered. • User discomfort minimization is not considered.
Mathematical programming	Multi-objective mixed integer linear programming [52]	• Cost minimization. • User comfort maximization. • PAR minimization.	• Load scheduling period.	• TOU	• Analytic hierarchy process based differential evolution algorithm	• Integration of RES and BESS is lacking.
Meta-heuristic algorithm	Beetle antennae search [53]	• Cost minimization. • Privacy protection.	• PEV, load and grid power constraints. • Minimum/maximum energy, charging/discharging rate, power and SoC of PEV. • Load scheduling period.	• Price-Based DR • Incentive-Based DR	N.A.	• Integration of RES is lacking. • PAR reduction is not considered. • User discomfort minimization is not considered. • Model implication for multiple homes is neglected.
Meta-heuristic algorithm	Modified butterfly optimization algorithm [54]	• Cost minimization. • User comfort maximization.	• PV system, BESS, load and grid power constraints. • Minimum/maximum value of capacitor in the BESS. • Load scheduling period.	• Price-Based DR	• Particle swarm algorithm • Butterfly optimization algorithm	• PAR reduction is not considered.
Meta-heuristic algorithm	Multi-Objective arithmetic optimization [55] algorithm	• Cost minimization. • User comfort maximization. • PAR minimization.	• Charging/discharging rate and maximum capacity of the battery. • Load scheduling period.	• RTP • Critical peak pricing	• Particle swarm optimization • Gray wolf optimizer • Antlion optimization	• PV system, load and grid power constraints are not taken into account.
Meta-heuristic algorithm	Cuckoo optimization algorithm [56]	• Cost minimization. • User comfort maximization. • PAR minimization.	• Load scheduling period.	• TOU	• Mixed integer linear programming	• Integration of RES and BESS is lacking. • Formulated objective function as weighted function.

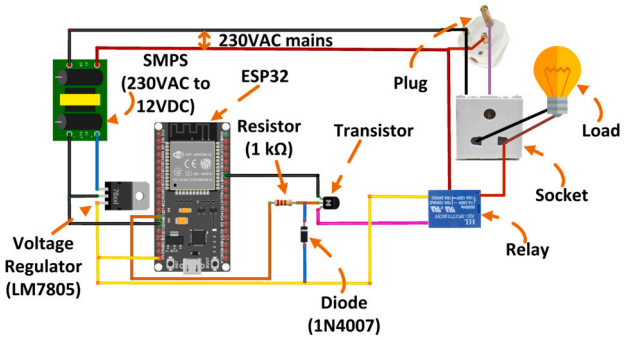


Fig. 3. Smart plug schematic diagram.

Utilizing the multi-objective mountain gazelle optimization approach, the PSC schedules appliance operations to minimize costs and maximize efficiency. The PSC then sends signals to users, indicating the optimal times to power on or off their appliances. Users review the PSC's schedule and authorize the scheduling of appliances they find convenient. The smart plug automates the appliance operation based on the authorized schedule, leveraging IoT for real-time data integration and execution.

3.2. Application of multi-objective optimization in load scheduling

The authors of the study adopted the multi-objective mountain gazelle optimization approach for load scheduling due to its rapid computation and nearly ideal outcomes, irrespective of the magnitude of the issue. This multi-objective optimization considers various factors such as cost minimization, Peak-to-Average Ratio (PAR) minimization, and user discomfort minimization. By balancing these objectives, the approach ensures efficient energy usage while maintaining user satisfaction. The algorithm's ability to handle multiple objectives simultaneously makes it highly suitable for complex load scheduling tasks in smart grids, where diverse and sometimes conflicting goals must be achieved.

3.3. Architecture of smart plug

Smart plugs are intermediary devices that connect household appliances to the HEMS. Each appliance is connected to a smart plug, enabling remote control. The circuit diagram of smart plug is shown in Fig. 3, where a relay is used within the socket to connect and disconnect the appliance from the mains. The paramount controlling element of the smart plug is the ESP32, which is responsible for receiving signals from the user and controlling the relay's on/off status. Since the ESP32 energises at 3.3 V, the 230 V ac supply is converted to 5 V using a switched-mode power supply board, and 5 V is then stepped down to 3.3 V using a voltage regulator. However, the digital on/off signals of ESP32 cannot be fed directly to the relay module as it does not provide sufficient driving current; hence, a transistor and a diode are added between them.

3.4. Implementation steps

Initially, users input their desired operating times for appliances through a mobile application. The PSC collects this data along with TOU pricing information from the utility and schedules appliance operations using the mountain gazelle optimization algorithm. The optimized schedule is then sent to users for approval, and they authorize the schedules for appliances they find convenient. The smart plug receives the final authorized schedule, and the ESP32 controls the relay to turn the appliance on or off at the scheduled times. The hybrid inverter ensures optimal use of PV, BESS, and grid power based on real-time availability and demand. Excess PV power is used to charge the BESS or is sent to the grid, while the grid power charges the BESS during low rates if PV power is insufficient.

4. Mathematical modeling of system building blocks

4.1. Regular and uninterruptible-time-deferrable device modeling

In this prototype, every enrolled dwelling $h \in H$ in a cluster has regular and uninterruptible-time-deferrable devices; the total appliance of a user is typified by set $D = D_{rl} \cup D_{ul}$. Regular devices (D_{rl}) are operated readily when intended, whereas uninterruptible-time-deferrable devices (D_{ul}) can be rescheduled for usage at low peak periods to attain an equalised load curve, respectively. The DR is executed at the commencement of each day, with each day being segmented into fixed-duration slots of one hour, for a total of 24 time slots. These time slots are denoted by $\tau \in T = \{1, 2, 3,..., 24\}$ and the ON/OFF status of both types of devices at τ is denoted by a binary decision variable $y^{\tau}_{h,d_{rl}}$, $y^{\tau}_{h,d_{ul}} \in \{0, 1\}$.

Each home h has a set of regular loads D_{rl} such that $d_{rl} \in D_{rl}$. Moreover, the power consumption of each regular appliance (d_{rl}) is denoted as $S_{h,d_{rl}}$, while the cumulative energy consumption of all regular appliances (D_{rl}) in a cluster is computed using the following equation:

$$E_{rl}^{\tau} = \sum_{h \in H} \sum_{d_{rl} \in D_{rl}} S_{h,d_{rl}} \times y_{h,d_{rl}}^{\tau} \tag{1}$$

Furthermore, each uninterruptible-time-deferrable load is symbolized as $d_{ul} \in D_{ul}$, while its power consumption is represented as $S_{h,d_{ul}}$. These loads operate under a user-defined scheduling window and cannot be suspended while running. Thus, the user-specified start time ($\tau_{begin_{h,d_{ul}}}$) and end time ($\tau_{end_{h,d_{ul}}}$) must be greater than the appliance's required working time $\left(\tau_{span_{h,d_{ul}}}\right)$. The total energy consumption of all uninterruptible-time-deferrable load (D_{ul}) in a cluster is governed by the equation:

$$E_{ul}^{\tau} = \begin{cases} \sum_{h \in H} \sum_{d_{ul} \in D_{ul}} S_{h,d_{ul}} \times y_{h,d_{ul}}^{\tau}, & \tau_{begin_{h,d_{ul}}} \le \tau \le \tau_{end_{h,d_{ul}}} \\ 0, \ otherwise \end{cases} \tag{2}$$

$$\tau_{begin_{h,d_{ul}}} \ge \tau_{s_{h,d_{ul}}} \tag{2a}$$

$$\tau_{begin_{h,d_{ul}}} + \tau_{span_{h,d_{ul}}} \le \tau_{end_{h,d_{ul}}} \le \tau_{e_{h,d_{ul}}} \tag{2b}$$

Where, τ_s is the primordial permissible start time and τ_e is the latter permissible finish time.

4.2. Power deferrable device modeling

The BESS serves as a standby entity for housing energy, and its charging and discharging strategies have a profound influence on system reliability, economic implications, and ecological dimensions. When the state of charge (SoC) of the battery approaches the minimum threshold, P_{bch}^{τ} is the amount of electric power (kW) imported by the battery during the time slot τ. The battery is charged either by PV when surplus power is available or by the grid when the electricity tariff is lower.

The BESS can perform charging or discharging within a time frame, which is expressed through a binary decision variable $y_{bt}^{\tau} \in \{0, 1\}$. The power exported from the battery to load while discharging is symbolised by P_{bdch}^{τ}. The charging and draining efficiencies of a battery are η_{bch} and η_{bdch}, respectively, and the amount of energy stored in the battery, E_{bt}^{τ}, is determined by Eq. (3).

$$E_{bt}^{\tau} = E_{bt}^{\tau-1} + P_{bch}^{\tau} \cdot \eta_{bch} \cdot y_{bt}^{\tau} \cdot \Delta\tau - \frac{P_{bdch}^{\tau} \cdot \left(1 - y_{bt}^{\tau}\right) \cdot \Delta\tau}{\eta_{bdch}} \tag{3}$$

As a provision to maintain the battery in a pristine state, deep draining of the battery should be prevented by limiting the depth of discharge (DoD_{bt}), and the energy stored in the battery should not surpass the maximum battery capacity ($E_{bt_{max}}^{\tau}$), as described in Eq. (3a). Furthermore, the charging and discharging power of the BESS should not exceed the manufacturer-approved discharge (RT_{dch}) and charge rates (RT_{ch}), as expressed in Eqs. (3b) and (3c).

$$E_{bt_{max}}^{\tau} \cdot \left(1 - DoD_{bt}\right) \le E_{bt}^{\tau} \le E_{bt_{max}}^{\tau} \tag{3a}$$

$$0 \le P_{bch}^{\tau} \le RT_{ch} \cdot y_{bt}^{\tau} \tag{3b}$$

$$0 \le P_{bdch}^{\tau} \le RT_{dch} \cdot \left(1 - y_{bt}^{\tau}\right) \tag{3c}$$

4.3. PV panel modeling

Each residence in the community has a rooftop solar panel system coupled with a maximum power point tracker installed to proficiently harvest and deliver power to appliances while distributing excess energy to the BESS or grid. As stated in Eq. (4), the proportion of solar irradiation (Ir^{τ}) striking the panel surface area (Ar_{pvm}) at an ambient temperature T_o^{τ} and its efficiency (η_{pvm}) to transform light energy into electrical energy are scrutinised to estimate the energy yielded by the PV system (SE_{pvm}^{τ}) per time slot τ.

$$SE_{pv}^{\tau} = Ir^{\tau} \cdot \eta_{pvm} \cdot Ar_{pvm} \cdot \left(1 - 0.005\left(T_o^{\tau} - 25\right)\right) \tag{4}$$

However, the gross solar energy (SE_{total}) produced in a day is harvested between dawn (τ_{du}) and dusk (τ_{da}) as stated in Eq. (5).

$$SE_{total} = \sum_{\tau=\tau_{da}}^{\tau=\tau_{du}} SE_{pv}^{\tau} \tag{5}$$

4.4. Energy management system modeling

The HEMS is interconnected with the grid, PV, BESS, and household appliances; thus, it must exert independent control over variables subject to power constraints at time slot τ, as shown in the Eq. (6) in order to establish power balance in the energy system.

$$E_{gbuy}^{\tau} + SE_{pv}^{\tau} + E_{bdch}^{\tau} = E_{rl}^{\tau} + E_{ul}^{\tau} + E_{bch}^{\tau} + E_{gsell}^{\tau} \tag{6}$$

Where, the energy available in the battery while charging (E_{bch}^{τ}) and discharging (E_{bdch}^{τ}) is defined as follows:

$$E_{bt}^{\tau} = \begin{cases} E_{bch}^{\tau}, & y_{bt}^{\tau} = 1 \\ E_{bdch}^{\tau}, & y_{bt}^{\tau} = 0 \end{cases} \tag{6a}$$

Moreover, E_{gbuy}^{τ} and E_{gsell}^{τ} depict the energy consumed and transferred to the grid, respectively, and are capped such that neither transaction exceeds the critical tolerable energy acquisitions ($E_{MAXgbuy}^{\tau}$) and sales ($E_{MAXgsell}^{\tau}$) imposed by the utility is indicated as follows:

$$0 \le E_{gbuy}^{\tau} \le E_{MAXgbuy}^{\tau} \tag{6b}$$

$$0 \le E_{gsell}^{\tau} \le E_{MAXgsell}^{\tau} \tag{6c}$$

5. Appliance scheduling criteria formulation

Objective function 1: Minimization of cost metrics

In this research, the fundamental scheduling requirement is to minimize the operational cost of the household appliances, as stated in Eq. (7), by adopting the TOU pricing structure, which is conventionally governed and conveyed to the user by the power grid operator.

$$Minimize\ f_{cost} = \sum_{\tau=1}^{T} \left(E_{rl}^{\tau} + E_{ul}^{\tau}\right) \times C_{TOU}^{\tau} \tag{7}$$

C_{TOU}^{τ} denotes the TOU tariff signal for time slot τ, which may vary seasonally and have multiple levels throughout the day.

Objective function 2: Attenuation of PAR metrics

A cluster of residences may experience drastic increase in peak during off-peak intervals if the first objective function is solely optimized. Hence, to avert this circumstance, an additional objective function defined in Eq. (8) is introduced that will reduce PAR while maintaining the dependability and stability of the grid.

$$Minimize\ f_{PAR} = \frac{\max\limits_{\tau \in T} \left(E_{rl}^{\tau} + E_{ul}^{\tau}\right)}{\frac{1}{T}\sum_{\tau=1}^{T} \left(E_{rl}^{\tau} + E_{ul}^{\tau}\right)} \tag{8}$$

Objective function 3: Attenuation of user-inconvenience metrics

The two objective functions outlined above address the reduction of peaks and expenses, which may result in an early or delayed commencement time for appliances that might interfere with end-user comfort. As a result, Eq. (9) is modeled for each home, with the scheduler offering a different start time ($\tau_{sch_{d_{ul}}}$) for an appliance instead of $\tau_{begin_{d_{ul}}}$; this delay is referred to as the appliance's waiting time ($\tau_{wait_{d_{ul}}}$).

$$Minimize\ f_{\tau_{wait_{d_{ul}}}} = \frac{\sum_{\mathrm{d_{ul}} \in D_{\mathrm{ul}}} \left|\tau_{begin_{d_{ul}}} - \tau_{sch_{d_{ul}}}\right|}{D_{\mathrm{ul}}} \tag{9}$$

However, the three objectives are minimized within the operating limits and constraints and is formulated as follows:

$$\begin{aligned} Minimize\ f \quad &= \left(f_{cost}, f_{PAR}, f_{\tau_{wait_{d_{ul}}}}\right) \\ s.t. \quad &constraints\ (2) - (6) \end{aligned} \tag{10}$$

6. Optimization strategy

6.1. Mountain gazelle optimization

In order to identify the ideal solution for the objective function delineated in the preceding section, the authors of the paper have fabricated the single-objective mountain gazelle optimization (MGO) [57] technique as a multi-objective mountain gazelle optimization (MMGO) method.

MGO is a nature-inspired, swarm-based optimization approach that utilises the social behavior of mountain gazelles to address practical engineering challenges. Initially, a random population is generated within the search space, as illustrated in Eq. (11), where each element of search space Y_{nd} represents the n-th gazelle of d-th dimension.

$$Y_{nd} = Y_{min} + rand\left(Y_{max} - Y_{min}\right) = \begin{bmatrix} Y_{1,1} & Y_{1,2} & \cdots & Y_{1,d} \\ Y_{2,1} & Y_{2,2} & \cdots & Y_{2,d} \\ \vdots & \vdots & \vdots & \vdots \\ Y_{n,1} & Y_{n,2} & \cdots & Y_{n,d} \end{bmatrix} \tag{11}$$

Where, $rand$ is a uniform random vector varying between 0 and 1, while Y_{min} and Y_{max} are the lower and upper limits of Y_{nd}. Subsequently, Eq. (11) is applied in objective function for estimating the fitness value of each population during optimization and subsequent matrix collect the fitness value of all population:

$$H_{pop} = \begin{bmatrix} f\left(Y_{1,1};\, Y_{1,2};\, \cdots;\, Y_{1,d}\right) \\ f\left(Y_{2,1};\, Y_{2,2};\, \cdots;\, Y_{2,d}\right) \\ \vdots \\ f\left(Y_{n,1};\, Y_{n,2};\, \cdots;\, Y_{n,d}\right) \end{bmatrix} \tag{12}$$

Where, H_{pop} is the matrix for saving the fitness of each population, and f is an objective function. Each solution of H_{pop} can become a member of one of the herds among territorial solitary males, maternity herds and bachelor male herds. However, the fittest solution obtained is called adult male gazelle (gaz_{male}). Further, the exploitation and exploration phases are performed in parallel using four mechanisms including territorial solitary males, maternity herds, bachelor male herds, and migration to search for food. The behavioral pattern of territorial solitary males involves trying to occupy other territory as well as protect their own, in which young males help them, which is mathematically modeled in Eq. (13).

$$TMH = gaz_{male} - |rd \times gaz_{male_{child}} - rd \times Y(t) \times B_1(d) \times e^{2-It\times\left(\frac{2}{M_It}\right)}| \times Coef_n \tag{13}$$

In Eq. (13), rd is a random integer of value 1 or 2 and B_z is a random number from the standard distribution for z = {1,2,3,4}. Using Eq. (15), $Coef_n$ is derived. To improve the search efficiency, one coefficient vector is randomly chosen from a set of four coefficient vectors with n = {1,2,3,4}. The young male herd coefficient vector ($gaz_{male_{child}}$) is computed using Eq. (14). M_It is the total number of iterations, and It is the current number of iterations. Also, $Y(t)$ is the position of the gazelle vector in the current iteration.

$$gaz_{male_{child}} = Y_a \times rand + L \times rand,\ a = \left\{\frac{n}{3} \cdots n\right\} \tag{14}$$

In Eq. (14), Y_a is a random solution lying in the interval of a. L is the mean of $\frac{1}{3}^{rd}$ of randomly selected gazelle in a search space and $rand$ are random values between 0 and 1.

$$Coef_n = \begin{cases} (It \times \left(\frac{-1}{M_It}\right)) + rand, \\ \left(-1 + It \times \left(\frac{-1}{M_It}\right)\right) \times B_2(d), \\ rand, \\ B_3(d) \times B_4(d)^2 \times \cos\left(rand \times 2 \times B_3(d)\right). \end{cases} \tag{15}$$

On the other hand, maternity herds have significance in gazelle gestation and this behavior is formulated using Eq. (16) where Y_{rand} is the vector position of a gazelle, which is chosen arbitrarily from the search space.

$$PH = gaz_{male_{child}} + Coef_n + (rd \times gaz_{male} - rd \times Y_{rand}) \times Coef_n \tag{16}$$

Subsequently, Eq. (17) is used to formulate the behavior of bachelor male herds where the adolescent gazelles participate in combat with adult male gazelles.

$$SMH = Y(t) - K + (rd \times gaz_{male} - rd \times gaz_{male_{child}}) \times Coef_n \tag{17}$$

Where,

$$K = (|Y(t)| + |gaz_{male}|) \times (2 \times rand - 1) \tag{18}$$

Furthermore, as shown in Eq. (19) the gazelles traverse far-flung regions for grazing and relocation, where $bound_{upper}$ and $bound_{lower}$ are the upper and lower limits, of objective function, respectively.

$$WFS = (bound_{upper} - bound_{lower}) \times rand + bound_{lower} \tag{19}$$

Hence, the MGO optimization algorithm performs an optimization procedure based on the life events of mountain gazelles, including TMH, PH, SMH, and WFS, with each gazelle belonging to one of the packs. However, TMH, PH, and SMH are responsible for producing offspring. At the completion of each iteration, robust gazelles with optimal solutions are saved and designated as adult male gazelles. Other solutions that do not produce satisfactory results are considered elderly and ailing gazelles and are eliminated from the search space.

Table 2
Appliance data used in this study.

Appliance type	Appliances	Power rating (kW)	Operational Period (hrs)
Uninterruptible-time-deferrable	Electric oven-1 (morning hours)	2.15	0.75
	Kettle	2	0.5
	Electric iron	1.5	0.5
	Vacuum cleaner	0.7	1
	Blender	0.3	0.25
	Washing machine with dryer	1.9	3
	Dish washer	1.8	2
	Electric oven-2 (evening hours)	2.15	1
Regular	Refrigerator	0.18	24
	Lightning	0.3	0.18 (6p.m -12a.m.)
	Fans (x3)	0.195	0.195 (1p.m. – 4p.m. and 6p.m. – 12 a.m.)
	Personal Computer	0.2	0.2 (1p.m. – 4p.m. and 6p.m. – 12 a.m.)
	Television	0.103	0.103 (8p.m. – 11p.m.)

6.2. Multi-objective mountain gazelle optimization

In order to achieve a balance between all the objective functions with the fewest possible trade-offs, the mountain gazelle optimization issue is remodeled into a multi-objective optimization problem as shown in Eq. (10). In multi-objective optimization techniques, no single optimal solution is often discovered; instead, either dominated ($\vec{\alpha}$) or non-dominated solutions ($\vec{\beta}$) are obtained. The Pareto optimal set alternatives with non-dominated solutions are considered viable solution spaces and are termed as Pareto front, as illustrated in Eq. (20).

$$F\left(\vec{\beta}\right) < F\left(\vec{\alpha}\right) \tag{20}$$

The optimal solution among the Pareto fronts is chosen using fuzzy set theory as shown in Eq. (21).

$$\epsilon_j^p = \begin{cases} 1, & F_j \le F_j^{min} \\ \frac{F_j^{min} - F_j}{F_j^{max} - F_j^{min}} & F_j^{max} < F_j < F_j^{min} \\ 0, & F_j \ge F_j^{max} \end{cases} \tag{21}$$

Where, F_j is the objective function and j is the number of objective functions with Pareto optimal solution p. The value of ϵ_j^p lies between 0 and 1, with the ideal solution being nearest to 1. According to Eq. (22), satisfaction of each Pareto solution is calculated and the solution with the greatest value is chosen as the optimal compromise solution.

$$\epsilon_j^p = \frac{\sum_{j=1}^{l} \epsilon_j^p}{\sum_{j=1}^{k} \sum_{j=1}^{l} \epsilon_j^p} \tag{22}$$

Where, k represents the total non-dominated solution and l represents the total objectives.

7. Input data

7.1. Functioning period and load characteristics of the appliances

Table 2 provides statistics on the operational time and load characteristics of the regular and uninterruptible-time-deferrable devices. As mentioned in Eqs. (2) and (3), users can specify the desired operating periods for their uninterruptible-time-deferred devices; unless no preferred functioning intervals are provided, the load will be scheduled at any time of day to minimize cost and PAR.

7.2. Electricity tariff

Based on the tariffs indicated in Table 3, the current study is conducted. The tariff is set by the Tamil Nadu Electricity Board (TNEB), which is a power generation and distribution company in the state of Tamil Nadu, owned by the Government of Tamil Nadu, India.

Table 3
ToU tariff scheme.

TNEB Price	Time of Day	Interval of Day	Rate (Rs/kWh)
TOU Tariff	06.00 am to 10.00 am 05.00 pm to 09.00 pm	Peak hours	10.2
	10.00 am to 05.00 pm	Normal hours	8.5
	09.00 pm to 06.00 am	Off Peak hours	8.075
Feed-in-Tariff (1-10 kW)	-	-	5.74

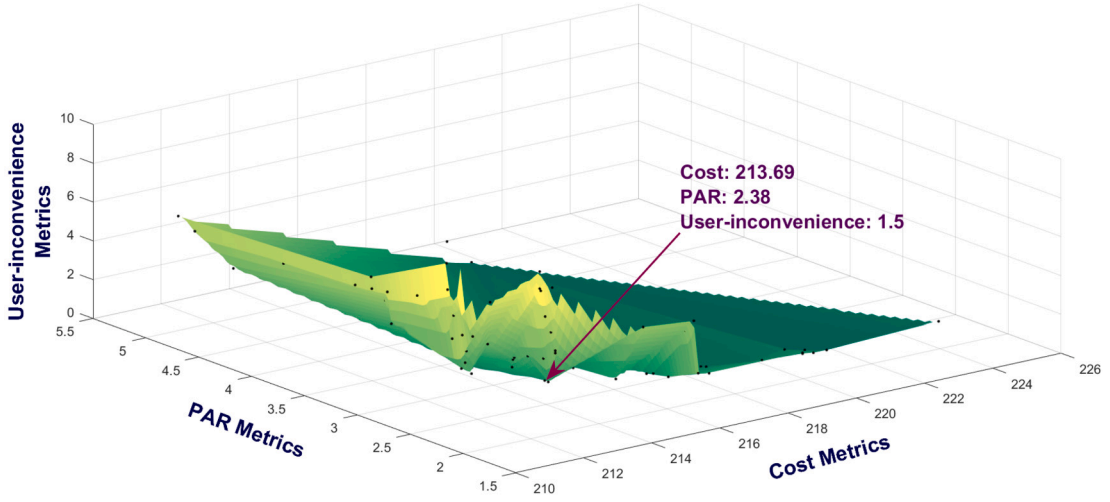


Fig. 4. Pareto front solutions for tri-objective using MMGO algorithm.

7.3. Photovoltaic array power output and BESS specifications

The estimated consumption for a single house, as indicated in Table 2, led to the selection of a rooftop PV system of 3.96 kW consisting of 12 numbers of 330 W Waaree panels and 16 lead-acid batteries of 150 Ahr. The PV data is extracted from Solcast [58], a company that offers relevant data on solar irradiation as well as the predicted power output of PV panels for each site based on the installed units. The information is updated globally every 10 to 15 minutes using numerical weather prediction models and pictures from third-generation weather satellites (Himawari8, GOES-16, and GOES-17).

8. Simulation results and discussion

This section compares the performance of the suggested multi-objective load scheduling model with that of existing multi-objective strategies including NSGA-II [35], MAOA [55], MFF [23], and single-objective constraint-based optimization methods like GWCSO [33] and CSA [56] to demonstrate the adequacy of the suggested methodology. In multi-objective load schedule modeling, cost, PAR, and user discomfort are classified as distinct objective functions. Whereas in a single-objective constraint-based optimization technique, the cost and user discomfort functions are presented as a single function using weights, and the PAR function is treated as a constraint.

For simulation analysis, a residential model is designed in which each home is equipped with appliances, as tabulated in Table 2. The user provides the preferred time to operate each appliance one day before and according to that the scheduling takes place using MMGO algorithm. Fig. 4 portrays the Pareto front dispersion and the best compromise alternative among the tri-objective functions designed for a single home. Additionally, the potential of PV and BESS integration is also investigated. The simulations are performed 30 times, and the average outcomes of each approach are contrasted.

8.1. Single dwelling without PV and BESS

For a scheduled load, the cost of purchasing power each day using the MAOA, CSA, NSGA-II, GWCSO, MFF, and MMGO optimization strategies is Rs. 214.51, Rs. 214.65, Rs. 214.31, Rs. 214.03, Rs. 214.26, and Rs. 213.39, respectively, with MMGO resulting in the most economical solution as illustrated in Fig. 5a, whereas an unplanned load costs Rs. 242.88. Hence, compared to unscheduled load, the cost reduction employing MAOA, CSA, NSGA-II, GWCSO, MFF, and MMGO for scheduled load is 11.68%, 11.62%, 11.76%, 11.87%, 11.78%, and 12.14% respectively, while the cost curtailment during peak hours is 77.45%, 76.09%, 78.43%, 77.60%, 78.52%, and 80.15% as depicted in Figs. 5a and 5b. The performance of scheduled and unscheduled loads in relation to PAR is displayed in Fig. 6. It has been noted that, in comparison to MAOA, CSA, NSGA-II, GWCSO, MFF, and unscheduled loads, the PAR of the suggested technique is, respectively, 1.56%, 21.74%, 3.63%, 32.61%, 21.74% and 52.54% less and demand curtailment during peak

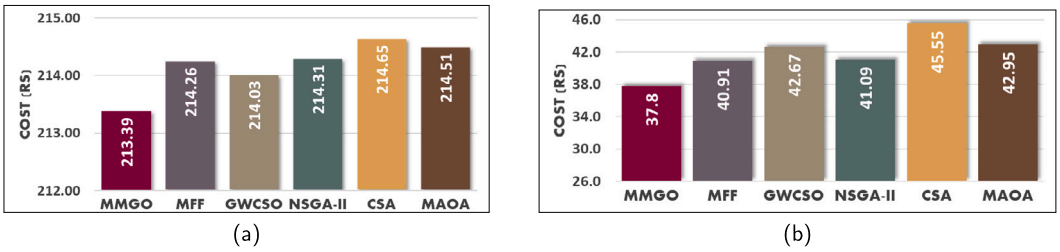


Fig. 5. Statistical cost evaluation of a home using the suggested and state-of-the-art techniques (a) throughout the day and (b) during peak hours.

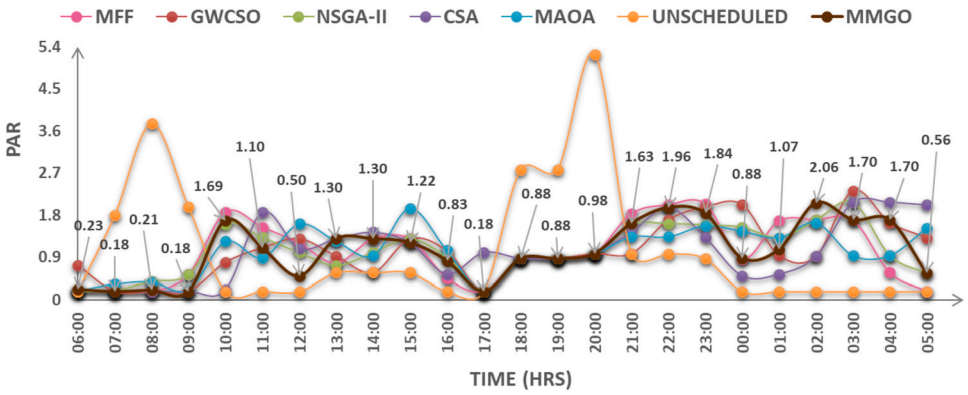


Fig. 6. Projecting hourly PAR profile using recommended and various other optimization methods.

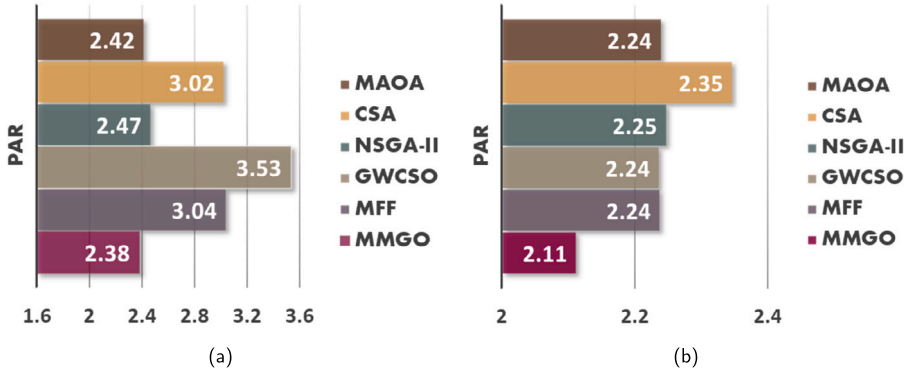


Fig. 7. PAR evaluation of a home (a) throughout the day (b) during peak hours using various strategies.

hours is 5.67%, 9.98%, 6.01%, 5.55%, 5.61%, and 25.07% as depicted in Figs. 7a and 7b. Correspondingly, in contrast to MAOA, CSA, NSGA-II, GWCSO, and MFF, respectively, the suggested method user discomfort index is 29.23%, 38.00%, 8%, 40.25%, and 38.66% less, as shown in Fig. 8. Although all algorithms considered in this study adhere to the objective functions and their constraints, the MMGO optimization technique demonstrated exceptional performance.

8.2. Multiple dwelling without PV and BESS

In the preceding section, the scheduling of a single household without coordination with the other households in the cluster was carried out, and its effect on cost, waiting time, and PAR were investigated through simulations. However, coordination between multiple HEMS is necessary when scheduling of appliances is carried out for a community since it would ease the strain on the local power network and prevent residential load shedding. Hence, a PSC scheduler is employed to schedule every load in the cluster at once. For simulation analysis, a cluster of 50 households with identical appliance data is considered and scheduled in accordance with the objectives outlined in Eqs. (7), (8), and (9). The results obtained demonstrate that the suggested technique lowers electricity bills by 2.83%, 9.44%, 4.3%, 6.73%, and 1% in comparison to MFF, GWCSO, NSGA-II, CSA, and MAOA, respectively, at peak hours. Subsequently, the proposed method exhibits excellent results by reducing overall PAR metrics to 3.4%, 65.91%, 7.1%, 15%, and 1%, in contrast to MFF, GWCSO, NSGA-II, CSA, and MAOA. Additionally, the MMGO method attained the lowest user discomfort metrics value of 1.27, whereas the MFF, GWCSO, NSGA-II, CSA, and MAOA approaches produced 1.5, 1.75, 1.7, 2.21, and 2.5, respectively.

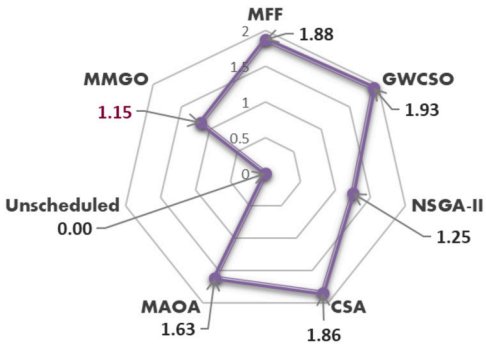


Fig. 8. User discomfort index using various methods.

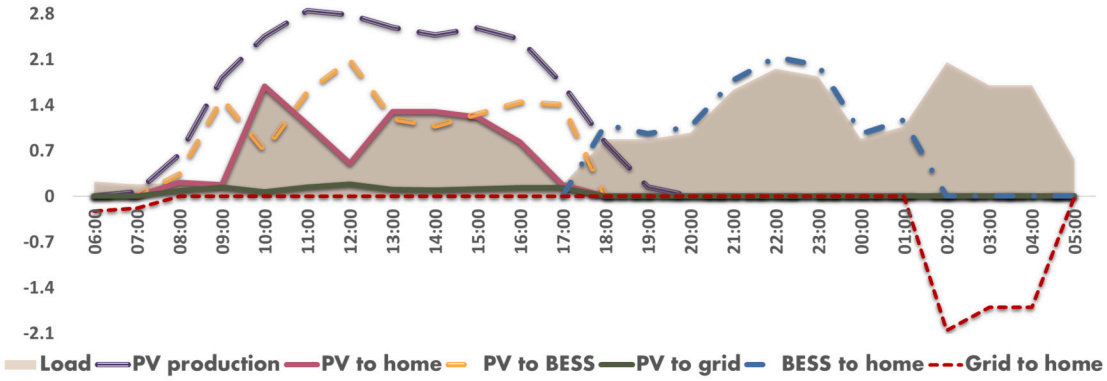


Fig. 9. 24-Hour energy production and consumption profile of a single residence.

8.3. HEMS with PV and BESS for a single dwelling

The preceding sections demonstrate that the proposed approach outperforms the existing methods. However, to further reduce the PAR and electricity costs, the integration of PV and BESS in proposed HEMS model is demonstrated. Hence, the capacity of PV and BESS described in section 7.3 is implemented on a single household consisting of total load as shown in Table 2. It is assumed that the SoC of battery is 30% at the start of day. The PV data for this study, sourced from Solcast, shows that the household generates 21.6365 kW of PV energy between 9:00 and 17:00, as illustrated in Fig. 9. During this period, the household reduces its dependence on the grid by utilizing 8.28 kW of PV power to fulfil house energy demands. The additional 12.24 kW of PV energy is used to charge the BESS, and the surplus electricity is sold back to the grid. Notably, the BESS discharge mainly occurs during the night, specifically between 18:00 and 1:00. The PAR and power cost reductions using the proposed method are up to 0.12% and 90%, respectively, in comparison to households without PV and BESS. Besides, the user comfort index has increased by 89.13%.

9. Technoeconomic analysis

Technoeconomic analysis aids in evaluating the benefits of operating infrastructure and provides insight into forthcoming capital investments. In this section, the cash payback period (CPP) is evaluated through the net present value (NPV_{HEMS}) of the HEMS system, as stated in Eq. (23). This helps to visualize the cash flow and the value of money over time, which would help investors with financial evaluation.

$$NPV = -(CC_{Smart\ Plug} + CC_{RES} + CC_{BESS}) + \sum_{j=1}^{yr} \frac{CF_j}{(1+rt)^j} \tag{23}$$

The NPV_{HEMS} comprises of initial investment and expected cash flow, where a positive NPV_{HEMS} indicates that the project is going to be profitable. The primary investment involves capital cost of smart plugs ($CC_{Smart\ Plug}$), PV system (CC_{RES}) and BESS system (CC_{BESS}) as illustrated in Table 4, whereas, the net cashflow (CF) for the total number of years (yr) of the project lifetime and the discount rate (rt), which are assumed to be 25 years and 8%, respectively, is the net cash flow (CF). The CF for using HEMS for yr year is estimated by deducting the operation and maintenance bills (OMB) from the monetary-savings on energy bills (EBS).

$$CF_{yr} = EBS_{yr} - OMB_{yr} \tag{24}$$

Table 4
Rating and price of materials.

Components	Rating	Initial cost (Rs)	Replacement cost (Rs)	OMB_{yr} (Rs/year)
PV Panels (x12)	330 W, 12 V	96480	96480	500
Galvanized iron structure	-	2000	-	-
AC and DC Distribution box, DC wire and MC4 connectors	-	9300	-	-
Lightning arrestor	-	1000	-	-
Earthing kit with insulation	-	4000	-	-
Installation charges	-	10000	-	-
Battery (x16)	150Ahr, 12 V	192000	192000	500
Inverter	5.25KVA, 48 V	110000	110000	1000
Smart Plug	-	1000	1000	-

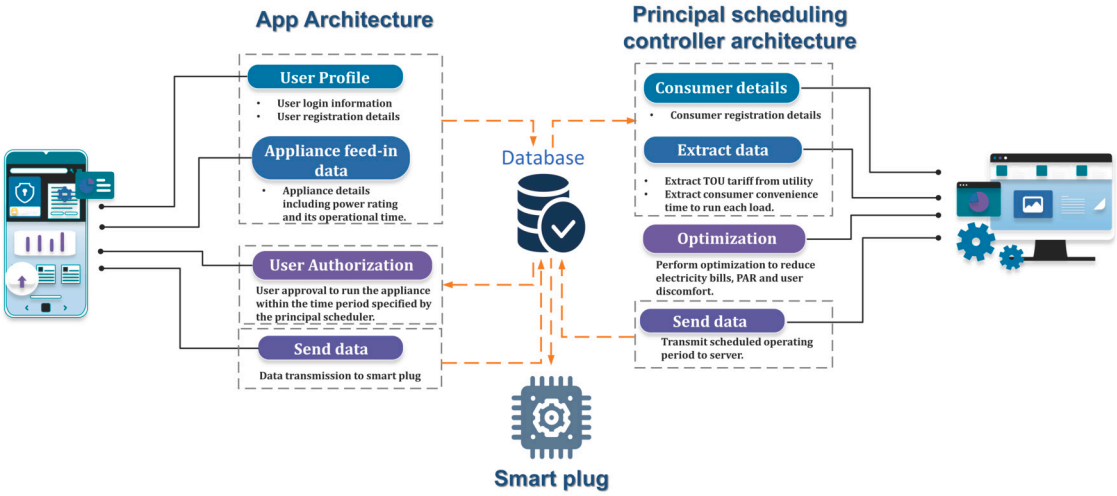


Fig. 10. The major design consideration of the proposed system.

However, the cash payback period as determined in Eq. (25) is the timeframe when NPV approaches zero, indicating that the capital invested is recovered, and beyond this point, the investor will be profitable. In this CPP analysis, 8% and 3% inflation rates are considered for electricity prices and HEMS operation and maintenance cost, respectively.

$$-\left(CC_{Smart\ Plug}+CC_{RES}+CC_{BESS}\right)+\sum_{j=1}^{yr}\frac{CF_j}{(1+rt)^j}=0 \tag{25}$$

Hence, the CPP for the proposed system found out to be 12 years which is 48% of project life time. However, from the case study, it is confirmed that the proposed scheme is practicable and profitable with any number of household appliances.

10. Hardware and software implementation

10.1. Development of mobile application

Initially, a user-friendly mobile app named Indravati is created using the MIT app inventor-2 with the intent of sending and receiving data from a database. As illustrated in Fig. 10, the app enables users to provide the PSC scheduler with their preferred scheduling time and receive the final scheduled time in return.

The app developed is password protected and has the feature where the user can feed the begin and commencement time of each appliance using start and stop time picker as depicted in Figs. 11b and 11c. The app takes care of whether the start and stop time entered by the user has a delay equivalent to operational time of appliance or else it displays error message as indicated in Fig. 11d.

The user of each household is allowed to enter and modify the data infinitely many times, from one day before until 23:30 hrs, after which the entered time will be locked. As seen in Fig. 11e, the Thingspeak cloud stores the user-desired appliance operating timings, which are then sent to the PSC scheduler. Further, the optimal time obtained by PSC scheduled is again stored in Thingspeak and displayed in user app within an interval of 15 seconds. If the user agrees to run the appliance at the scheduled time, they can choose through the checkbox as depicted in Fig. 11f, and the data will be sent to the smart plug. The authorized start time is then saved in a real-time database, with which the smart plug will be linked and continually access data. Thus, with a maximum delay

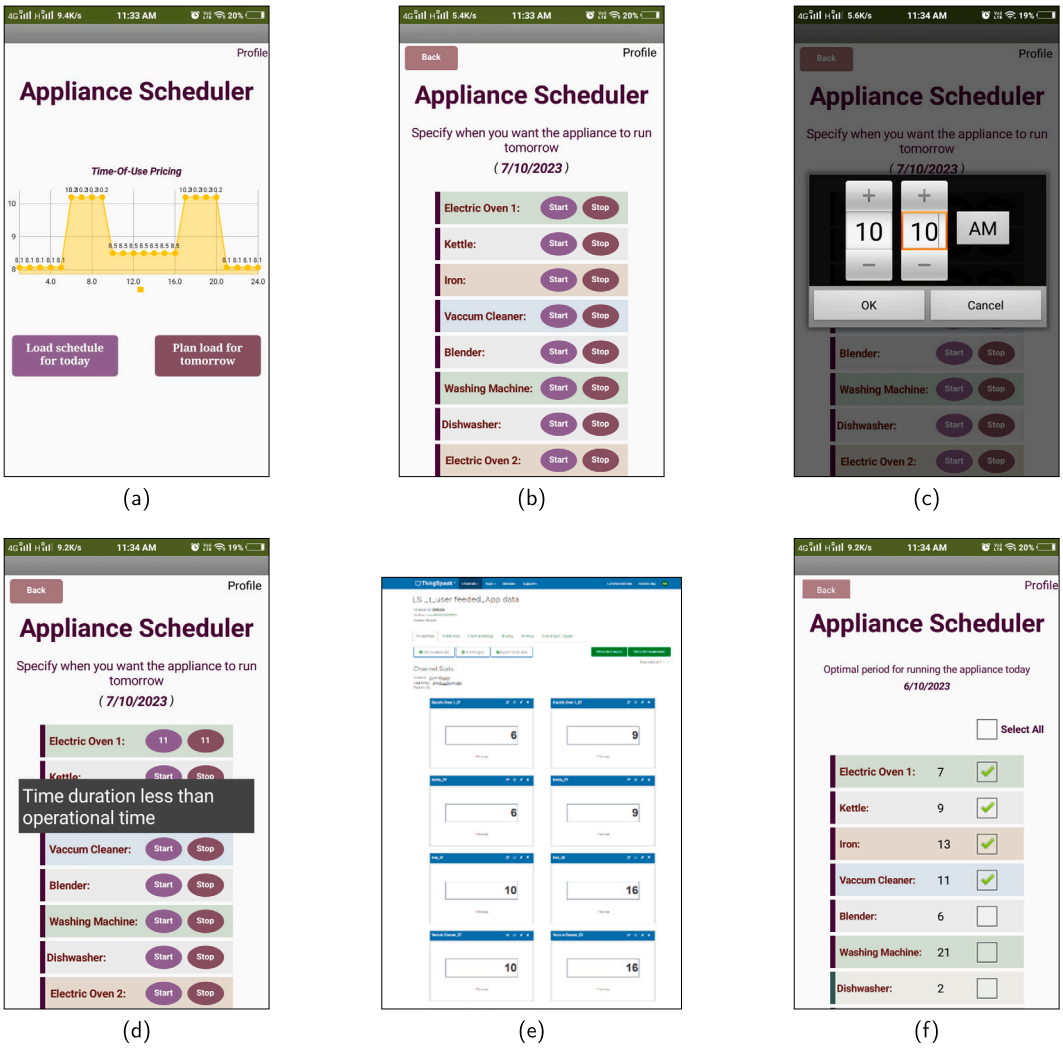


Fig. 11. (a) Indravati mobile app interface, (b) mobile app interface to set appliance begin and end time period, (c) time picker and (d) error message displayed when stop time is less than load operating profile length, (e) data stored in ThingSpeak, and (f) Scheduled time received from principal scheduling controller.

of one second, the smart plug is activated at the predetermined time. Moreover, the app also provides the feature of adding new appliance where the consumer needs to enter the name, power rating and operating time period of appliance.

10.2. Development of PSC standalone desktop application

A PSC scheduler is designed using MATLAB graphic user interface (GUI) which allows the development and deployment of standalone desktop applications. As shown in Fig. 12 the scheduler has the appliance data of all individual households in a cluster and performs scheduling between 23:00 hrs to 00:00 hrs. The desktop app initially extracts data from Thingspeak and performs scheduling using MMGO algorithm and reverts back to Thingspeak with scheduled start time as depicted in Fig. 10.

10.3. Development of smart plug

The smart plug is implemented in hardware as shown in Fig. 13 in accordance with the specifications provided in Section 3.3, where ESP32 is the brain of smart plug. The ESP32 receives the scheduled time for individual appliance through Indravati App. Currently, the model is developed for one appliance; however, the ESP32 has 18 analog-to-digital converter pins through which each appliance can be connected to each channel. So, by employing one ESP32 for one room the proposed system can be made much more economical.

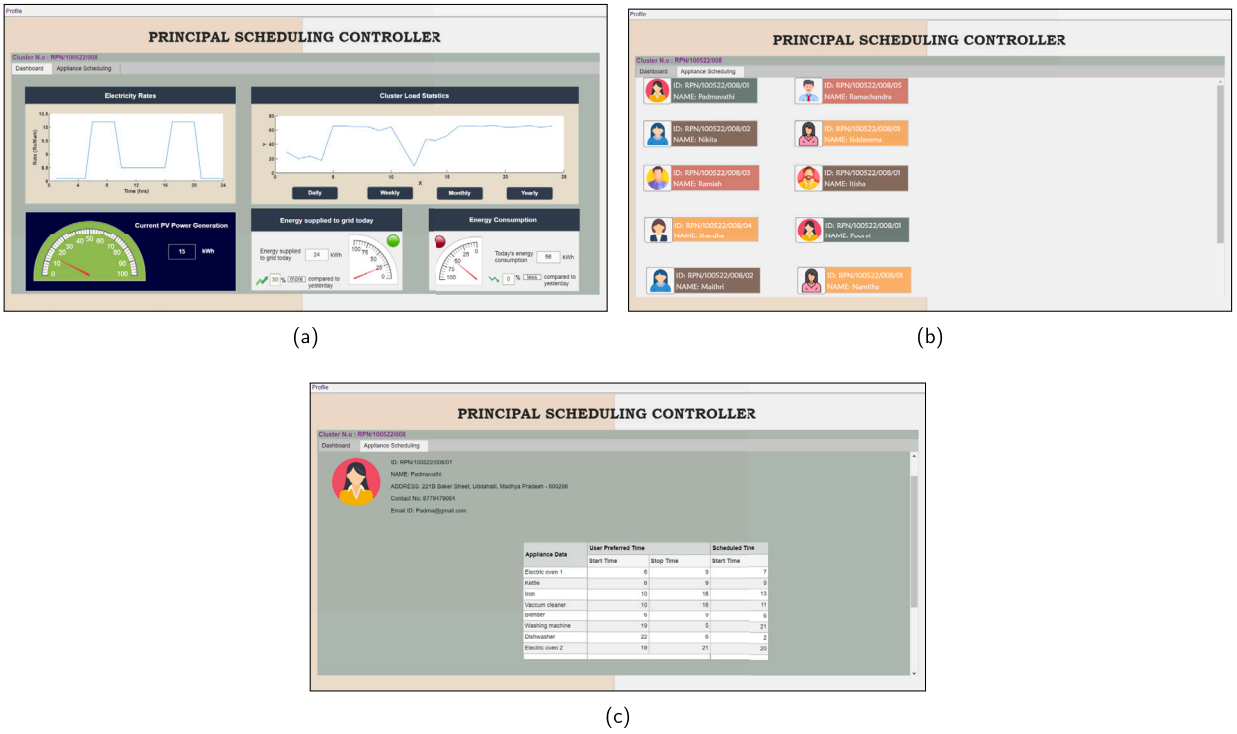


Fig. 12. (a) Principal scheduling controller dashboard, (b) user profile of each home participating in DR program, and (c) optimized start time generated for each appliances.

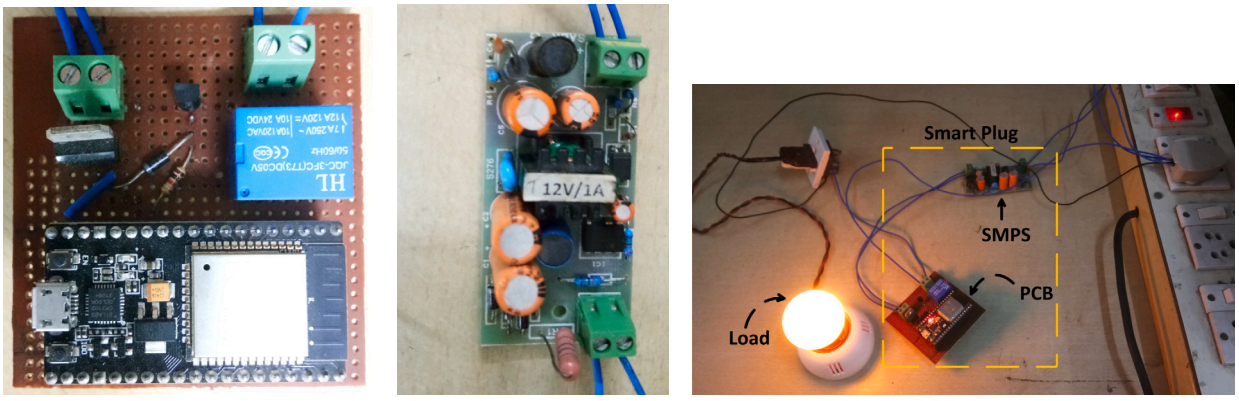


Fig. 13. (a) PCB of a smart plug, (b) switched mode power supply board, and (c) real-time execution of the entire system.

11. Conclusion

This study demonstrates a systematic methodology for developing HEMS blocksets/architecture across multiple residences. The components of HEMS are explained in detail, and each system element is integrated through firmware and hardware solutions.

- A mobile application named Indravati serves as a front-end interface, designed for the purpose of data collection, which is essential for feeding the PSC.
- The PSC software is developed for the utility to schedule loads for all HEMS in the cluster.
- The PSC optimizes the energy usage by employing Multi-Objective Mountain Gazelle Optimization (MMGO) algorithm, which extends the MGO method with archive maintenance and fuzzy logic-based pareto selection. The performance of proposed optimization method is contrasted with five existing strategies. The key findings for a single-home scenario include:
  - Cost Reduction: The MMGO method effectively reduced electricity bills by 7.66%, 11.47%, 8.06%, 17.07%, 12.05%, and 80.15% during peak hours when compared to existing methods like MFF, GWCSO, NSGA-II, CSA, and MAOA.

  - PAR Reduction: The MMGO method has shown significant improvements in PAR metrics, with a reduction of 2.64 compared to unscheduled methods. In contrast, the MFF, GWCSO, NSGA-II, CSA, and MAOA methods achieved PAR reductions of 1.98, 1.49, 2.55, 2.00, and 2.60, respectively.
  - User Comfort: The incorporation of PV and BESS within the Home Energy Management System (HEMS) for a single dwelling led to up to a 90% reduction in power costs and a 0.12% reduction in PAR, along with an 89.13% increase in user comfort index.
- Technoeconomic analysis is performed on the proposed system, which reveals that the proposed system, when integrated with PV and BESS, has a cash payback period (CPP) of 12 years, constituting 48% of the project lifetime. This demonstrates the financial viability and long-term benefits of the system.

Future research could significantly improve this study by conducting studies on user behavior and preferences, which will help refine the load scheduling algorithms to better align user comfort and convenience with energy savings. Additionally, testing the scalability of the proposed system in larger residential communities (like cities) and various geographical locations can provide insights into its adaptability and effectiveness across different environments.

CRediT authorship contribution statement

Nikita Ramachandra: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Rajasekar Natarajan:** Writing – review & editing, Writing – original draft, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India, Energy Statistics India 2023, Energy Statistics India, 2023, https://mospi.gov.in/publication/energy-statistics-india-2023, 2023. (Accessed 8 October 2023), Online.
[2] M. Asif, Handbook of Energy Transitions, CRC Press, 2022.
[3] N. Venkatesan, J. Solanki, S.K. Solanki, Residential demand response model and impact on voltage profile and losses of an electric distribution network, Appl. Energy 96 (2012) 84–91, https://doi.org/10.1016/j.apenergy.2011.12.076, smart Grids.
[4] S. Lin, L. Zhao, F. Li, Q. Liu, D. Li, Y. Fu, A nonintrusive load identification method for residential applications based on quadratic programming, Electr. Power Syst. Res. 133 (2016) 241–248, https://doi.org/10.1016/j.epsr.2015.12.014.
[5] X. Wu, X. Hu, X. Yin, C. Zhang, S. Qian, Optimal battery sizing of smart home via convex programming, Energy 140 (2017) 444–453, https://doi.org/10.1016/j.energy.2017.08.097.
[6] K.M. Tsui, S.C. Chan, Demand response optimization for smart home scheduling under real-time pricing, IEEE Trans. Smart Grid 3 (4) (2012) 1812–1821, https://doi.org/10.1109/TSG.2012.2218835.
[7] S. Aslam, A. Khalid, N. Javaid, Towards efficient energy management in smart grids considering microgrids with day-ahead energy forecasting, Electr. Power Syst. Res. 182 (2020) 106232, https://doi.org/10.1016/j.epsr.2020.106232.
[8] A. Rochd, A. Benazzouz, I. Ait Abdelmoula, A. Raihani, A. Ghennioui, Z. Naimi, B. Ikken, Design and implementation of an ai-based & iot-enabled home energy management system: a case study in Benguerir — Morocco, in: Technologies and Materials for Renewable Energy, Environment and Sustainability, Energy Rep. 7 (2021) 699–719, https://doi.org/10.1016/j.egyr.2021.07.084.
[9] J. Yang, Q. Sun, L. Yao, Y. Liu, T. Yang, C. Chu, L. Fang, L. Zhu, A novel dynamic load-priority-based scheduling strategy for home energy management system, J. Clean. Prod. 389 (2023) 135978, https://doi.org/10.1016/j.jclepro.2023.135978.
[10] S.L. Arun, M.P. Selvan, Intelligent residential energy management system for dynamic demand response in smart buildings, IEEE Syst. J. 12 (2) (2018) 1329–1340, https://doi.org/10.1109/JSYST.2017.2647759.
[11] Y.-J. Kim, A supervised-learning-based strategy for optimal demand response of an hvac system in a multi-zone office building, IEEE Trans. Smart Grid 11 (5) (2020) 4212–4226, https://doi.org/10.1109/TSG.2020.2986539.
[12] E. Mocanu, D.C. Mocanu, P.H. Nguyen, A. Liotta, M.E. Webber, M. Gibescu, J.G. Slootweg, On-line building energy optimization using deep reinforcement learning, IEEE Trans. Smart Grid 10 (4) (2019) 3698–3708, https://doi.org/10.1109/TSG.2018.2834219.
[13] D. Azuatalam, W.-L. Lee, F. de Nijs, A. Liebman, Reinforcement learning for whole-building hvac control and demand response, Energy AI 2 (2020) 100020, https://doi.org/10.1016/j.egyai.2020.100020.
[14] R. Lu, S.H. Hong, M. Yu, Demand response for home energy management using reinforcement learning and artificial neural network, IEEE Trans. Smart Grid 10 (6) (2019) 6629–6639, https://doi.org/10.1109/TSG.2019.2909266.
[15] M. Alizadeh Bidgoli, A. Ahmadian, Multi-stage optimal scheduling of multi-microgrids using deep-learning artificial neural network and cooperative game approach, Energy 239 (2022) 122036, https://doi.org/10.1016/j.energy.2021.122036.
[16] O. Samuel, N. Javaid, A. Khalid, W.Z. Khan, M.Y. Aalsalem, M.K. Afzal, B.-S. Kim, Towards real-time energy management of multi-microgrid using a deep convolution neural network and cooperative game approach, IEEE Access 8 (2020) 161377–161395, https://doi.org/10.1109/ACCESS.2020.3021613.
[17] R. Tang, H. Li, S. Wang, A game theory-based decentralized control strategy for power demand management of building cluster using thermal mass and energy storage, Appl. Energy 242 (2019) 809–820, https://doi.org/10.1016/j.apenergy.2019.03.152.
[18] G. Aghajani, H. Shayanfar, H. Shayeghi, Demand side management in a smart micro-grid in the presence of renewable generation and demand response, Energy 126 (2017) 622–637, https://doi.org/10.1016/j.energy.2017.03.051.
[19] M. Hu, F. Xiao, Price-responsive model-based optimal demand response control of inverter air conditioners using genetic algorithm, Appl. Energy 219 (2018) 151–164, https://doi.org/10.1016/j.apenergy.2018.03.036.
[20] B. Mota, P. Faria, Z. Vale, Residential load shifting in demand response events for bill reduction using a genetic algorithm, Energy 260 (2022) 124978, https://doi.org/10.1016/j.energy.2022.124978.

[21] P. Carrasqueira, M.J. Alves, C.H. Antunes, Bi-level particle swarm optimization and evolutionary algorithm approaches for residential demand response with different user profiles, Inf. Sci. 418–419 (2017) 405–420, https://doi.org/10.1016/j.ins.2017.08.019.
[22] G. Chen, X. Yi, Z. Zhang, H. Wang, Applications of multi-objective dimension-based firefly algorithm to optimize the power losses, emission, and cost in power systems, Appl. Soft Comput. 68 (2018) 322–342, https://doi.org/10.1016/j.asoc.2018.04.006.
[23] Y. Yang, J. Qiu, Z. Qin, Multidimensional firefly algorithm for solving day-ahead scheduling optimization in microgrid, J. Electr. Eng. Technol. 16 (2021) 1755–1768, https://doi.org/10.1007/s42835-021-00707-7.
[24] S.N. Makhadmeh, A.T. Khader, M.A. Al-Betar, S. Naim, A.K. Abasi, Z.A.A. Alyasseri, A novel hybrid grey wolf optimizer with min-conflict algorithm for power scheduling problem in a smart home, Swarm Evol. Comput. 60 (2021) 100793, https://doi.org/10.1016/j.swevo.2020.100793.
[25] J. Zhang, Y. Wu, Y. Guo, B. Wang, H. Wang, H. Liu, A hybrid harmony search algorithm with differential evolution for day-ahead scheduling problem of a microgrid with consideration of power flow constraints, Appl. Energy 183 (2016) 791–804, https://doi.org/10.1016/j.apenergy.2016.09.035.
[26] B.N. Silva, K. Han, Mutation operator integrated ant colony optimization based domestic appliance scheduling for lucrative demand side management, Future Gener. Comput. Syst. 100 (2019) 557–568, https://doi.org/10.1016/j.future.2019.05.052.
[27] B. Lu, N. Razmjooy, A modified fractional-order-based future search algorithm for performance enhancement of a pemfc-based cchp, Energy Sources, Part A: Recov. Utiliz. Environ. Effects 45 (2023) 12821–12843, https://doi.org/10.1080/15567036.2023.2276385.
[28] N. Javaid, M. Naseem, M.B. Rasheed, D. Mahmood, S.A. Khan, N. Alrajeh, Z. Iqbal, A new heuristically optimized home energy management controller for smart grid, Sustain. Cities Soc. 34 (2017) 211–227, https://doi.org/10.1016/j.scs.2017.06.009.
[29] M.S. Javadi, M. Gough, M. Lotfi, A. Esmaeel Nezhad, S.F. Santos, J.P. Catalão, Optimal self-scheduling of home energy management system in the presence of photovoltaic power generation and batteries, Energy 210 (2020) 118568, https://doi.org/10.1016/j.energy.2020.118568.
[30] K. Murugaperumal, P.D.V. Raj, Integrated energy management system employing pre-emptive priority based load scheduling (pepls) approach at residential premises, Energy 186 (2019) 115815, https://doi.org/10.1016/j.energy.2019.07.145.
[31] E.S.F. Tantawy, G.M. Amer, H.M. Fayez, Scheduling home appliances with integration of hybrid energy sources using intelligent algorithms, Ain Shams Eng. J. 13 (4) (2022) 101676, https://doi.org/10.1016/j.asej.2021.101676.
[32] M. Waseem, Z. Lin, S. Liu, Z. Zhang, T. Aziz, D. Khan, Fuzzy compromised solution-based novel home appliances scheduling and demand response with optimal dispatch of distributed energy resources, Appl. Energy 290 (2021) 116761, https://doi.org/10.1016/j.apenergy.2021.116761.
[33] M. Waseem, Z. Lin, S. Liu, I.A. Sajjad, T. Aziz, Optimal gwcso-based home appliances scheduling for demand response considering end-users comfort, Electr. Power Syst. Res. 187 (2020) 106477, https://doi.org/10.1016/j.epsr.2020.106477.
[34] Z. Yahia, A. Pradhan, Optimal load scheduling of household appliances considering consumer preferences: an experimental analysis, Energy 163 (2018) 15–26, https://doi.org/10.1016/j.energy.2018.08.113.
[35] A.H. Sharifi, P. Maghouli, Energy management of smart homes equipped with energy storage systems considering the par index based on real-time pricing, Sustain. Cities Soc. 45 (2019) 579–587, https://doi.org/10.1016/j.scs.2018.12.019.
[36] Q. Lu, Z. Zhang, S. Lü, Home energy management in smart households: optimal appliance scheduling model with photovoltaic energy storage system, Energy Rep. 6 (2020) 2450–2462, https://doi.org/10.1016/j.egyr.2020.09.001.
[37] B. Lokeshgupta, S. Sivasubramani, Multi-objective home energy management with battery energy storage systems, Sustain. Cities Soc. 47 (2019) 101458, https://doi.org/10.1016/j.scs.2019.101458.
[38] J. Zhu, Y. Lin, W. Lei, Y. Liu, M. Tao, Optimal household appliances scheduling of multiple smart homes using an improved cooperative algorithm, Energy 171 (2019) 944–955, https://doi.org/10.1016/j.energy.2019.01.025.
[39] D. Wang, X. Zhang, K. Qu, T. Yu, Z. Pan, Q. Liu, Pareto tribe evolution with equilibrium-based decision for multi-objective optimization of multiple home energy management systems, Energy Build. 159 (2018) 11–23, https://doi.org/10.1016/j.enbuild.2017.10.072.
[40] T. Sattarpour, D. Nazarpour, S. Golshannavaz, A multi-objective hem strategy for smart home energy scheduling: a collaborative approach to support microgrid operation, Sustain. Cities Soc. 37 (2018) 26–33, https://doi.org/10.1016/j.scs.2017.09.037.
[41] Q. Lu, S. Lü, Y. Leng, Z. Zhang, Optimal household energy management based on smart residential energy hub considering uncertain behaviors, Energy 195 (2020) 117052, https://doi.org/10.1016/j.energy.2020.117052.
[42] H. Golmohamadi, R. Keypour, B. Bak-Jensen, J. Radhakrishna Pillai, Optimization of household energy consumption towards day-ahead retail electricity price in home energy management systems, Sustain. Cities Soc. 47 (2019) 101468, https://doi.org/10.1016/j.scs.2019.101468.
[43] A. Mohammadi Rad, T. Barforoushi, Optimal scheduling of resources and appliances in smart homes under uncertainties considering participation in spot and contractual markets, Energy 192 (2020) 116548, https://doi.org/10.1016/j.energy.2019.116548.
[44] I. Gonçalves, Álvaro Gomes, C. Henggeler Antunes, Optimizing the management of smart home energy resources under different power cost scenarios, Appl. Energy 242 (2019) 351–363, https://doi.org/10.1016/j.apenergy.2019.03.108.
[45] B. Chreim, M. Esseghir, L. Merghem-Boulahia, Losish—load scheduling in smart homes based on demand response: application to smart grids, Appl. Energy 323 (2022) 119606, https://doi.org/10.1016/j.apenergy.2022.119606.
[46] H.R. Rocha, I.H. Honorato, R. Fiorotti, W.C. Celeste, L.J. Silvestre, J.A. Silva, An artificial intelligence based scheduling algorithm for demand-side energy management in smart homes, Appl. Energy 282 (2021) 116145, https://doi.org/10.1016/j.apenergy.2020.116145.
[47] A. Bouakkaz, A.J.G. Mena, S. Haddad, M.L. Ferrari, Efficient energy scheduling considering cost reduction and energy saving in hybrid energy system with energy storage, J. Energy Storage 33 (2021) 101887, https://doi.org/10.1016/j.est.2020.101887.
[48] I.O. Essiet, Y. Sun, Z. Wang, Optimized energy consumption model for smart home using improved differential evolution algorithm, Energy 172 (2019) 354–365, https://doi.org/10.1016/j.energy.2019.01.137.
[49] M.S. Javadi, A.E. Nezhad, P.H. Nardelli, M. Gough, M. Lotfi, S. Santos, J.P. Catalão, Self-scheduling model for home energy management systems considering the end-users discomfort index within price-based demand response programs, Sustain. Cities Soc. 68 (2021) 102792, https://doi.org/10.1016/j.scs.2021.102792.
[50] A.R. Jordehi, Binary particle swarm optimisation with quadratic transfer function: a new binary optimisation algorithm for optimal scheduling of appliances in smart homes, Appl. Soft Comput. 78 (2019) 465–480, https://doi.org/10.1016/j.asoc.2019.03.002.
[51] M. Martinez-Pabon, T. Eveleigh, B. Tanju, Optimizing residential energy management using an autonomous scheduler system, Expert Syst. Appl. 96 (2018) 373–387, https://doi.org/10.1016/j.eswa.2017.12.017.
[52] Z. Yahia, A. Pradhan, Multi-objective optimization of household appliance scheduling problem considering consumer preference and peak load reduction, Sustain. Cities Soc. 55 (2020) 102058, https://doi.org/10.1016/j.scs.2020.102058.
[53] S. Zhang, J. Rong, B. Wang, An optimal scheduling scheme for smart home electricity considering demand response and privacy protection, Int. J. Electr. Power Energy Syst. 132 (2021) 107159, https://doi.org/10.1016/j.ijepes.2021.107159.
[54] X. Wang, X. Mao, H. Khodaei, A multi-objective home energy management system based on Internet of Things and optimization algorithms, J. Build. Eng. 33 (2021) 101603, https://doi.org/10.1016/j.jobe.2020.101603.
[55] D. Bahmanyar, N. Razmjooy, S. Mirjalili, Multi-objective scheduling of iot-enabled smart homes for energy management based on arithmetic optimization algorithm: a node-red and nodemcu module-based technique, Knowl.-Based Syst. 247 (2022) 108762, https://doi.org/10.1016/j.knosys.2022.108762.
[56] A. Shaban, H. Maher, M. Elbayoumi, S. Abdelhady, A cuckoo load scheduling optimization approach for smart energy management, Energy Rep. 7 (2021) 4705–4721, https://doi.org/10.1016/j.egyr.2021.06.099.
[57] B. Abdollahzadeh, F.S. Gharehchopogh, N. Khodadadi, S. Mirjalili, Mountain gazelle optimizer: a new nature-inspired metaheuristic algorithm for global optimization problems, Adv. Eng. Softw. 174 (2022) 103282, https://doi.org/10.1016/j.advengsoft.2022.103282.
[58] J.M. Bright, Solcast: validation of a satellite-derived solar irradiance dataset, Sol. Energy 189 (2019) 435–449, https://doi.org/10.1016/j.solener.2019.07.086.